

NCA Assistant

User Manual

Version 1.8

Non-Compartmental Pharmacokinetic Analysis

Bioequivalence • Drug Interaction Studies • Power & Sample Size

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Not validated for your use: qualifying your installation is your responsibility (see Intended Use and Your Data).

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Intended Use and Your Data

NCA Assistant is for pharmacokineticists doing non-compartmental analysis, average bioequivalence testing and study planning. It gives no reference-scaled bioequivalence verdict (ABEL, RSABE), and it has no audit trail, electronic signature or access control. The public instance on shinyapps.io is for evaluation and training, with synthetic or pseudonymised data. For regulated work, install a tagged release on your own system and qualify it there with the validation package. Responsibility for the analysis and its conclusions stays with the user. Not for dosing decisions for individual patients.

Your data. On the public instance, uploads are processed on shinyapps.io servers run by Posit PBC (USA). Upload only synthetic, example or anonymised data there.

Pseudonymised trial data are still personal data under the GDPR. Sending them to a third-party host needs agreements your organisation must have in place, and may breach sponsor confidentiality. For real study data, run the app on your own computer.

Status. NCA Assistant is not validated for your use. The validation package in the repository (a URS, an IQ/OQ/PQ protocol and a test script with more than 400 tests) is a starting point: the developer's test run shows that the code does what the tests check, on the developer's computer. Qualifying the installation you use is your responsibility; the chapter on regulated use explains how.

Supplier, support and warranty. The app is free, open-source software under the GNU General Public License, version 3 or later, written by Rob ter Heine (Radboud Applied Pharmacometrics, Radboudumc, Nijmegen). It comes without any warranty and without a support agreement. Problems and questions go to the issue tracker at github.com/Robterheine/NCAassistant/issues.

What the App Does Not Do

Use other software for the following. The app refuses some of these, but not all, so check before you start.

Not supported	What to use or do instead
Reference-scaled bioequivalence verdicts (EMA ABEL, FDA RSABE) and the FDA NTID scaled test	Software that implements them, for example the replicateBE package for ABEL. The app shows CVwR and the limits it would imply for information only.
Studies with three or more treatments in one analysis (for example a three-way food-effect or DDI crossover, a Williams design)	Analyse two treatments at a time (see Studies with Three or More Treatments), knowing the residual then comes only from those periods, or use software that fits the full design.
Dose-proportionality statistics (the power model)	Use the dose-normalised parameters and plots here, and fit the power model elsewhere.
Absolute or relative bioavailability between routes or strengths in one comparison	Run the NCA per route with dose normalisation and compute F outside the app.
Urine data (amount excreted, renal clearance, excretion rate)	Other NCA software.
Sparse or destructive sampling (one sample per subject, Bailer's method)	Other NCA software, or a population approach.

Therapeutic drug monitoring data with irregular dosing, superposition, several infusion durations in one run, baseline-corrected endogenous compounds	A population PK model, or other NCA software.
Dosing decisions for individual patients	Not an intended use.

Getting Started

Opening the Public App

Go to robterheine.shinyapps.io/NCAassistant in Chrome, Firefox, Edge or Safari. There is nothing to install. The public instance is for evaluation and training: read *Intended Use and Your Data* before you upload anything. It disconnects after a period without activity, and it keeps nothing between sessions.

Installing the App on Your Own Computer

For confidential data and for regulated work, run the app on your own computer. It needs R 4.1 or newer; RStudio is optional but makes the steps easier.

1. Install R from cran.r-project.org, and optionally RStudio Desktop from posit.co.
2. Download a release from github.com/Robterheine/NCAassistant/releases: under Assets, choose "Source code (zip)", and unzip it. Use a tagged release rather than the latest code, so you know which version you run.
3. In RStudio, open `install_and_run.R` from the unzipped folder and click Source. Without RStudio, open a terminal in that folder and run `Rscript install_and_run.R`.
4. The first time, the script installs about twenty R packages from CRAN, which can take several minutes. Later starts take a few seconds.
5. The app opens in your web browser. To stop it, press Esc in the RStudio console, or Ctrl+C in the terminal.

For a qualified installation, add `--validated` (`Rscript install_and_run.R --validated`), or run `NCA_VALIDATED <- TRUE` before sourcing the script. It then installs the package versions the release was validated with, listed in `validation/renv.lock`, instead of the latest ones from CRAN. Run it from the folder that contains `app.R`.

If the installation fails. Read the last lines the script printed: they name the package that failed. On Windows, a package that needs compiling may need Rtools (cran.r-project.org/bin/windows/Rtools); on a Mac, the Xcode command-line tools (`xcode-select --install` in a terminal). Then run the script again; packages that installed are not installed twice.

The Hub Screen

The hub has six cards, numbered in the order of a typical study. Plan a Study needs no data. Upload & Check Data is where every analysis starts; the other four paths use the data processed there.

Card	Needs uploaded data	What you get
1. Plan a Study	No	Sample size or power, with plots
2. Upload & Check Data	(starts here)	A quality report and the processed data for paths 3 to 6
3. Visualize Data	Yes	Concentration-time figures and a Figure Record

4. One Subject at a Time	No (manual entry) or yes	NCA of one profile, with the half-life inspector
5. All Subjects (Batch)	Yes	NCA of every profile, summary statistics, a Complete Analysis Record
6. Bioequivalence	Yes	NCA, ANOVA, confidence intervals, a verdict and a Complete Analysis Record

Once data are processed, the navigation bar shows a badge such as "12 subjects, 132 obs". Home, in the bar under the navigation, brings you back to the hub. Processing a new file clears every result of the previous one, and a result is also cleared when you change a setting it depends on: run the analysis again after a change. Settings stay when you leave a page and come back.

Getting Help Inside the App

Every setting that needs explaining has an ⓘ button next to it; click it for a short explanation in plain language. The top navigation bar has three more sources. The Data Guide shows how to lay out the data for each study type, with example files to download. Statistical Methods describes every calculation, with references, and ends with Example Methods Paragraphs to adapt for a report or thesis. About lists the packages, their versions and licences, and the version history.

Saving Your Work

The app keeps nothing between sessions. Before you close the browser tab, or leave the public app idle, download what you need: the results as Excel or CSV, the figure, or the Complete Analysis Record, which holds the data, the settings and a script that reproduces the analysis. After a disconnect, upload the file again and click Process Data.

Tutorials

Seven short tutorials, from a first analysis without a file to a bioequivalence study with partial AUCs. Each gives the steps with the labels the app shows, then questions. The answers are in the Answer Key at the end of this chapter, rounded as the app displays them. The example files are in the Data Guide (top navigation bar), under Example Datasets. The first time, do the tutorials in order: later ones assume you know the upload steps.

Tutorial 1: A First Analysis Without a File

1. On the hub, click 4. One Subject at a Time.
2. Under **Where is your data?**, choose **Enter data manually (type values directly)**. The Time points and Concentrations boxes already hold an example profile: eight samples from 0 to 24 h, one value per line.
3. Click **Use This Data**.
4. Under **Analysis Settings**, keep Dose 100, Dose unit mg, Time unit h, Conc unit ng/mL and **Route of administration Oral / IM / SC (extravascular)**. Click **Run PK Analysis**.
5. Read the **PK Parameters** card, then the **Half-Life Calculation** card, which shows the points used for the terminal phase.

Questions: 1) What are Cmax and Tmax? 2) What is the AUC to last point? 3) What is the half-life, and on how many points does it rest? 4) What is the apparent clearance CL/F, and in which unit?

Tutorial 2: Uploading Data with BLQ Results

This tutorial uses example_blq.csv: eight subjects after a 100 mg oral dose, with results below the limit of quantification written as "<0.5".

1. On the hub, click 2. Upload & Check Data. Keep **What kind of file?** at **Simple table (one row per sample)**, and choose example_blq.csv.
2. Check the column mapping: Subject ID is Subject, Time is Time, Concentration is Concentration.
3. Leave **LLOQ value** at 0 and click **Process Data**. The app stops: it found BLQ text but no LLOQ. It shows "LLOQ auto-detected: 0.5" with a button **Set LLOQ to 0.5**.
4. Click **Set LLOQ to 0.5**. The button only fills in the field; check that it reads 0.5, keep **BLQ handling rule** at the default (Rule 1), and click **Process Data** again. The quality report now lists the 22 BLQ entries as a warning.
5. Go to 5. All Subjects (Batch). Keep **Dose information** at **Same dose for all subjects**, enter 100 as **Dose given to each subject**, and check the units (mg, h, ng/mL).
6. Under **Partial AUCs (optional)**, set **Number of intervals** to 1, with Start 0 and End 1. Click **Run Analysis on All Subjects**.

Questions: 1) Which subjects have a lag time, and how long is it (Lag Time, in **All Parameters (per profile)**)? 2) Subject 5 has a BLQ result at 8 h between two measurable ones. What is its time of last measurable concentration? 3) What is subject 2's Partial AUC 0–1? Now go back to Upload & Check Data, choose Rule 6 (Before first quantifiable →

LLOQ/2, rest → zero), click **Process Data**, and run the batch analysis again. What is subject 2's Partial AUC 0–1 now, and why did it change while the lag time did not?

Tutorial 3: All Subjects at Once (Theophylline)

1. Download example_theoph.csv and upload it. Map Subject ID to Subject, Time to Time and Concentration to conc. The app also maps the Dose column, which holds mg/kg. Leave **LLOQ value** at 0 and click **Process Data**; the badge shows 12 subjects, 132 obs.

2. Go to 5. All Subjects (Batch). Because a Dose column is mapped, **Dose information** is set to **Doses differ by subject or period (from Dose column in data)**. Switch it to **Same dose for all subjects** and enter 320 as **Dose given to each subject**, with Dose unit mg, Conc unit mg/L and Time unit h.

3. Keep **Route of administration** at **Oral / IM / SC (extravascular)** and **Trapezoidal method** at **Linear-up / Log-down**. Click **Run Analysis on All Subjects**.

Questions, with where to look: 1) Mean Cmax (**Summary Statistics**, Arithmetic Mean)? 2) Geometric mean AUC to last point (Summary Statistics, Geometric Mean)? 3) Median Tmax? 4) Which subject has the longest half-life, and how long is it (**All Parameters (per profile)**, sort by Half-Life)? 5) Geometric CV of AUC to infinity (observed) (Summary Statistics, Geometric CV (%))? 6) Is that longest half-life reliable? Tick **Show all parameters** and look at Points Used for Half-Life and AUC % Extrapolated (observed). 7) The note above the results names a subject with a pre-dose concentration above 5% of Cmax. Which subjects have a pre-dose concentration above zero, and what does that mean?

Tutorial 4: Bioequivalence

1. Download example_be_crossover.csv and upload it. Map Subject ID, Time and Concentration, and also Treatment / Formulation to Treatment, Period to Period and Sequence to Sequence. Click **Process Data**.

2. Go to 6. Bioequivalence. In **Step 1: PK Analysis Settings**, enter Dose 100 with units mg, h and ng/mL, and keep the route at **Oral / IM / SC (extravascular)**.

3. In **Step 2: Bioequivalence Settings**, check that **Study design** is **2×2 standard crossover (TR | RT)**, that **Reference treatment** is Reference, and that **Statistical model** is **Fixed effects (all terms fixed)**. Keep **Log-transform (recommended)**, the **Parameters to compare** (Cmax and AUC to last point), **Confidence interval** 90 and the limits 80 and 125.

4. Click **Run Complete BE Analysis**.

Questions: 1) What are the ratio and 90% CI for Cmax (tab **Test vs. Reference Comparison**, columns Estimate and 90% CI Lower and Upper)? 2) And for AUC to last point? 3) Is the product bioequivalent for both, and what does the note above the table say? 4) What is the within-subject CV for Cmax? It is not in the table on screen: click **Download CI Table (CSV)** and take the Residual Variance (MSE), then $CV = 100 \cdot \sqrt{(\exp(MSE) - 1)}$; or open Plan a Study, where a button offers the CV of this analysis.

Tutorial 5: Planning a Study

1. On the hub, click 1. Plan a Study.

2. Step 1: What do you want to calculate?: How many subjects do I need? (find sample size). Step 2: What kind of study?: Standard bioequivalence (80–125% limits, any region). Step 3: Study design: 2×2 standard crossover (TR | RT).

3. Under Parameters, enter **Within-subject variability (CV %)** 25, **Expected Test/Reference ratio (%)** 95 and **Target power** 80. Click **Calculate**.

Questions: 1) How many subjects are needed? 2) Switch Step 1 to **What is the chance of success for my planned study? (calculate power)**, enter 30 as **Total number of subjects in your study**, and click **Calculate**. What is the power? 3) Switch back to the sample size, set the CV to 35 and click Calculate. How many subjects are needed now? A result is cleared when you change an input: click Calculate again after every change.

Tutorial 6: Partial AUCs

Use the crossover data of Tutorial 4 (sampled at 0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8, 12 and 24 h).

1. In 5. All Subjects (Batch), enter Dose 100 (mg, h, ng/mL). Under **Partial AUCs (optional)**, set **Number of intervals** to 2. Interval 1: Start 0, End 1, and tick **Also report Cmax and Tmax in this interval**. Interval 2: Start 4, and tick **End at the last measurable concentration (t)**. Click **Run Analysis on All Subjects**.

2. Go to 6. Bioequivalence, with the settings of Tutorial 4. Each path has its own intervals, so enter them again in its Partial AUCs panel, with the Cmax box ticked for interval 1. Set **Role in bioequivalence** to **Pivotal (verdict)** for interval 1 and **Supportive (no verdict)** for interval 2. The partial metrics appear under **Parameters to compare**, already ticked. Click **Run Complete BE Analysis**.

Questions: 1) Work out Partial AUC 0–1 of subject 1 under Test by hand, from the first four samples (0, 12.3, 45.6 and 89.2 ng/mL), and compare it with the batch table. 2) What are the ratio and 90% CI for Partial AUC 0–1? 3) What does the table show for Partial AUC 4–t, and why? 4) What is the ratio for Cmax 0–1?

Tutorial 7: Figures

These three examples use Visualize Data (card 3). No analysis settings are needed.

A quick look after upload. Upload example_theoph.csv (Tutorial 3) and open 3. Visualize Data. The individual profiles plot shows 12 lines, one per profile, on a linear axis. Switch **Y-axis scale** to Log₁₀: a straight tail points to mono-exponential elimination.

A summary plot for a crossover. Upload example_be_crossover.csv with the Treatment column mapped (Tutorial 4), open Visualize Data and choose **Summary Plot**. It overlays one geometric mean curve per treatment. This is a visual check, not a test: only the confidence interval in Bioequivalence says whether the products are bioequivalent. For a report, go to **Export Figure** in the sidebar, choose PDF, 7 × 5 in and 300 DPI, and click **Download Figure**.

Dose-normalised profiles. With example_theoph.csv, whose Dose column differs per subject, tick **Normalise to dose (show C/Dose)** and colour the lines by Subject. Change the Y-axis label to "C/Dose (mg·L⁻¹ per mg/kg)". With dose-proportional pharmacokinetics the normalised profiles overlap.

Answer Key

Tutorial 1. 1) C_{max} 28.3 ng/mL at 1 h. 2) 156.8 ng/mL·h. 3) 4.607 h, from the last 5 points (adjusted R^2 0.9724). 4) 616.7 L/h: 100 mg divided by an AUC to infinity of 162.2 ng/mL·h, converted to litres.

Tutorial 2. 1) Subjects 2 and 5 (0.5 h), and 4 and 7 (0.25 h). 2) 24 h: under Rule 1 the BLQ result at 8 h becomes 0, but the measurable values after it still count, and the fall to zero adds its area to AUC. 3) 2.238 ng/mL·h under Rule 1 and 2.394 under Rule 6. Rule 6 sets the BLQ samples during the lag to LLOQ/2, which adds area before the first measured concentration. The lag time stays 0.5 h, because a value set by a BLQ rule is not a measured concentration.

Tutorial 3. 1) 8.759 mg/L. 2) 98.65 mg/L·h. 3) 1.135 h. 4) Subject 1, 14.3 h. 5) 28.43%. 6) Not very: it rests on 3 points, and 31.5% of AUC to infinity is extrapolated (flagged in amber). Theophylline's half-life is usually about 8 h; report this value with a flag, or leave it out of the descriptive statistics as your analysis plan says. 7) Subjects 1 (0.74 mg/L, 7% of its C_{max}), 7 and 10. Drug before the dose means carry-over from earlier dosing or an endogenous level; in a bioequivalence study ICH M13A excludes a period with a pre-dose value above 5% of C_{max} from the primary analysis. The single dose of 320 mg is also not right for every subject: subject 9 received $3.10 \text{ mg/kg} \times 86.4 \text{ kg} = 268 \text{ mg}$, so its CL/F is about 19% too high.

Tutorial 4. 1) 108.62% (106.27–111.02%). 2) 111.25% (108.95–113.60%). 3) Yes, both intervals lie within 80.00–125.00%. The note says that the study has 6 evaluable subjects, fewer than the 12 ICH M13A asks for, so the verdict is a statistical result only: this study would not be accepted as a pivotal study. 4) The residual variance is 0.000315, so the CV is about 1.8%. The example data are simulated with very little noise.

Tutorial 5. 1) 28 subjects. 2) 83.4%. 3) 52 subjects.

Tutorial 6. 1) 42.48 ng/mL·h ($1.5375 + 7.2375 + 33.7$); the concentrations rise throughout, so both trapezoidal methods give the same value. 2) 113.55% (110.90–116.25%), bioequivalent. 3) 112.54% (110.10–115.05%) without a verdict, because the interval is supportive. 4) 110.69% (108.31–113.11%).

Preparing Your Data and Study Types

General Format Rules

The app expects data in **long format**: one row per sample, with a subject identifier, the sampling time and the measured concentration. Accepted files are CSV (.csv), text (.txt, .tsv) and Excel (.xlsx, .xls). The app suggests a column mapping from the column names (Appendix C); check it before you process the data. When another column could also be the time or the concentration, a note names it. CDISC ADNCA datasets are uploaded differently, see Scenario 9.

Time is the time since the dose of that profile, with the pre-dose sample at 0. A negative pre-dose time is used as it is: the lag time then becomes negative and AUC includes the time before the dose, so set the pre-dose sample to 0.

Actual or nominal times. Use the actual sampling times for the NCA, as ICH M13A (2.2.2.2) asks: a sample taken at 1.1 h enters the calculation at 1.1 h. Nominal times (1 h) belong in the mean plot of Visualize Data, which averages per time point. Keep a second copy of the file with nominal times for that figure.

Decimal Separators and Regional Formats

Choose the delimiter (comma, semicolon or tab) and the decimal point (period or comma) in the upload settings. With a decimal comma, values such as 12,5 are read correctly, also in a concentration column that contains BLQ text such as “<0,5”. Excel files need no settings, because Excel stores numbers as numbers. In a decimal-comma file a point can only separate thousands, so 12.500 is read as 12500. Any other point in a number is refused, and the message names the column.

Units and Doses

Use one concentration unit, one time unit and one dose unit for the whole file, and select them in the app. They set the conversion factor for clearance and volume, so a wrong unit gives wrong CL/F and Vz/F while C_{max} and AUC look normal. A unit column with more than one unit (even ng/mL and µg/L) is refused; convert the values first. Molar concentrations (for example nmol/L) with a mass dose (mg) need the molecular weight, which you enter in the analysis settings. When the file states its units in a unit column (such as ConcUnit or Time_Unit, or the CDISC variables AVALU and RRLTU), the app pre-selects them and refuses a selection that contradicts them.

A Dose column must hold the amount given, in the dose unit you select. A weight-based dose in mg/kg with the unit set to mg gives clearance and volume per kilogram, labelled L/h and L. Convert mg/kg to mg first (dose × body weight), or report the values as per kilogram.

Scenario 1: Single Subject

To upload a file, include a Subject column; it may hold the same value on every row. Without it the upload cannot tell which rows belong together, and the app picks another column as Subject. Manual entry in One Subject at a Time needs only Time and Concentration (Tutorial 1).

Subject	Time	Concentration
---------	------	---------------

P1	0	0
P1	0.5	4.2
P1	1	18.5
P1	2	12.1
P1	4	6.3
P1	8	2.8
P1	24	0.4

Scenario 2: Multiple Subjects, Same Dose

Subject	Time	Concentration
S001	0	0
S001	1	22.5
S001	24	0.4
S002	0	0
S002	1	28.1
S002	24	0.6

Scenario 3: Multiple Subjects, Different Doses

Add a Dose column if subjects received different doses (dose escalation, weight-based dosing), or if a subject received different doses in different periods. The app takes the dose of each profile (subject, treatment and period) from this column, so every period is analysed with its own dose. For dose proportionality, see Scenario 11.

Scenario 4: Standard 2-Period Crossover (BE)

Six columns: Subject, Treatment, Period, Sequence, Time, Concentration. Time restarts at 0 in each period: it is the time since the dose of that period, not since the first dose of the study (the app refuses profiles that do not start near time zero). The ratio is Test ÷ Reference. In Bioequivalence you choose which treatment is the Reference; the app pre-selects it only when the name is unambiguous (R, Ref, Reference, Comparator, Innovator, Originator or RLD). Map the Period column. Without it a crossover is still analysed but gets no verdict, because the period effect cannot be separated from the treatment effect. Columns named APERIOD, Occasion or OCC are recognised as Period. A Visit column is not, so map it yourself if it holds the period. Subject IDs must be unique in the whole study: if each sequence numbers its subjects from 1, different people would be merged into one subject, so the app refuses the file.

If the protocol asks for a partial AUC, take a sample at every cutoff time. A cutoff between two samples is interpolated, so its value then depends on the trapezoidal method. The interval itself comes from the protocol or the product-specific guidance, never from the data.

Subject	Treatment	Period	Seq	Time	Conc
S01	Test	1	TR	0	0

S01	Test	1	TR	1	24.5
S01	Reference	2	TR	0	0
S01	Reference	2	TR	1	22.8

Scenario 5: Replicate Designs and Paired Comparison

Same six-column format. Replicate designs: 2×2×3 full replicate (sequences TRT and RTR), 2×3×3 partial replicate (TRR, RTR, RRT) and 2×2×4 full replicate (TRTR, RTTR). Period goes to 3 or 4 and a treatment appears more than once per subject; each administration is analysed as its own profile. Paired comparison: all subjects received the treatments in the same order, so Sequence has one value (or can be left out).

Scenario 6: Parallel-Group BE

Four columns: Subject, Treatment, Time, Concentration. No Period or Sequence needed.

Scenario 7: Steady-State PK

Same layout as Scenario 2, with one dosing interval per subject. Time 0 is just before the dose of that interval, and the pre-dose concentration is above zero. Include the pre-dose sample and sample up to the end of the dosing interval (τ). Tick “Steady-state (drug given repeatedly)” and enter τ in the app. If a profile has no pre-dose sample at time 0, the app assumes 0 there and says so, because AUC_{τ} is then too low.

Two concentrations describe the low point of the interval. C_{min} is the lowest observed concentration from 0 to τ , the pre-dose sample included; with an absorption lag it can fall after the dose. C_{τ} is the observed concentration at τ , and is left empty when no sample was taken at τ . ICH M13A lists both for steady-state studies, and the EMA modified-release guideline asks for C_{τ} . Both can be compared in Bioequivalence. AUC to infinity, its % extrapolated and the volume at steady state are single-dose quantities and are left empty.

A partial AUC at steady state has to lie within 0 and τ . AUC_{τ} is completed to τ with λ_z when the last sample comes earlier; a partial AUC you enter yourself is not, so it stays empty for such a profile.

Scenario 8: Drug–Drug Interaction Study

A drug interaction (DDI) study measures the effect of a co-administered drug (the perpetrator) on the pharmacokinetics of the victim drug. Most are crossovers in which each subject receives the victim drug alone and with the perpetrator, or fixed-order studies in which the victim is given alone first and with the perpetrator after a washout.

Use the same six columns as a crossover BE study (Scenario 4). Label the victim drug alone as “Reference” and the victim drug with the perpetrator as “Test”, or choose the Reference treatment in the app. The ratio is then (with perpetrator) ÷ (alone), the fold-change in exposure. For a single-dose victim, add AUC to infinity to the parameters to compare, as ICH M12 (5.1.1) asks; at steady state compare AUC_{τ} , C_{max} and C_{min} (ICH M12, 5.1.1). Analysing Drug–Drug Interaction Studies, in the chapter on using the app, explains how to read the result.

Scenario 9: CDISC ADNCA Dataset

If your data come as a CDISC ADNCA dataset (columns such as USUBJID, AVAL, PARAMCD, NRRLT, ARRLT and ANL01FL), do not convert them by hand. The steps:

1. On the Upload page, set What kind of file? to CDISC ADNCA dataset and upload the .csv or .xlsx file.
2. Read the summary, and choose the time to use: ARRLT or MRRLT (actual times) for the NCA, NRRLT (nominal times) for a mean figure; AFRLT, time since the first dose, is refused. With several analytes or matrices, choose the one to analyse; Set pre-dose times (negative ARRLT) to 0 does what it says.
3. Click Convert dataset. The app applies ANL01FL, drops records with PCSTAT "NOT DONE", copies PCLLOQ into the LLOQ field, and refuses datasets it cannot convert safely, such as derived records (DTYPE), several units, or AVAL missing without a BLQ result.
4. Check the column mapping and the LLOQ, then click Process Data.

The choices are stored in the Analysis Record. SAS transport (.xpt) files are not read: converters/ADNCA_TO_FLAT.md in the repository gives a two-line R recipe to convert them to CSV. The same conversion is available outside the app as converters/adnca_to_flat.R.

Scenario 10: IV Bolus and IV Infusion

For an IV bolus, choose IV Bolus (injected into vein at once) as the route. The concentration at time 0 is C0, the value just after the injection, not a pre-dose value: the app sets aside samples at or before time 0 and back-extrapolates C0 from the first two concentrations when they decline. Clearance and volume are reported without /F. AUC % Back-Extrapolated shows how much of AUC lies before the first sample; above about 20% the AUC is unreliable.

For an IV infusion, time 0 is the start of the infusion, and you enter the Infusion duration in the time unit of the data. One duration applies to every profile in a run, so split a file with infusions of different lengths and analyse each part separately. MRT is corrected for the infusion time (minus half the duration). Bioequivalence does not take an infusion.

Scenario 11: Single Ascending Dose and Dose Proportionality

Give every subject a Dose column and put the dose level in a Treatment column (for example "10 mg", "30 mg", "100 mg"). All Subjects (Batch) then gives summary statistics per dose group. Tick Calculate dose-normalised parameters to get Cmax, AUC and the partial AUCs divided by each profile's dose, and use Visualize Data with Normalise to dose (show C/Dose) to compare the profiles. Overlapping dose-normalised profiles suggest dose proportionality but do not test it. The power model ($\ln AUC = a + \beta \cdot \ln \text{Dose}$, with a confidence interval for β ; Smith et al., 2000) is not in the app; fit it with other software on the exported table.

Scenario 12: Multiple Dosing and Accumulation

A multiple-dose file usually holds a first-dose profile (Day 1) and a steady-state profile (Day N) for every subject. Code the day as Period, with time reset to 0 at each of the two doses,

and the day also in a Treatment column (for example "Day 1" and "Day 7"). The steady-state setting applies to every profile in a run, so:

1. For the Day N parameters, analyse the Day N profiles as steady state, with τ .
2. For the accumulation ratio, use Bioequivalence with Paired comparison, Day N as Test and Day 1 as Reference, Steady-state ticked with the same τ , and compare AUC Within Dosing Interval. Day 1 then gives AUC from 0 to τ after the first dose, and the ratio with its confidence interval is the accumulation ratio. The note about a missing pre-dose sample does not apply to Day 1.
3. Document that steady state was reached with consecutive trough samples, as ICH M13A (2.2.2.2) asks.

Scenario 13: Food Effect

A two-way food-effect study is a 2×2 crossover: the fed administration as Test and the fasted one as Reference (Scenario 4). Compare C_{max}, AUC to last point and AUC to infinity, and report the median difference in T_{max}. A 90% confidence interval within 80.00–125.00% supports the absence of a food effect. A three-way study (for example Test fasted, Test fed and Reference fasted) has three treatments, see Studies with Three or More Treatments.

Scenario 14: Metabolites and Several Analytes

Analyse one analyte per run. Filter a flat file to one analyte first, or choose it during the ADNCA upload. A metabolite-to-parent ratio needs both in molar units: enter the molecular weight of each in its own run, and compute the ratio from the exported tables.

Scenario 15: Absolute and Relative Bioavailability

Bioequivalence compares two treatments with one route and without dose normalisation, so it cannot compare IV with oral dosing, or strengths with different doses. Run All Subjects (Batch) per route with dose normalisation, and compute $F = (\text{AUC oral} / \text{dose oral}) \div (\text{AUC IV} / \text{dose IV})$ from the exported table. For a confidence interval of F, use other software.

Studies with Three or More Treatments

Bioequivalence takes exactly two treatments. For a three-way crossover (food effect, a three-arm DDI study, a Williams design), filter the file to two treatments per comparison and analyse each pair on its own. The residual variance then comes only from the periods of those two treatments, not from the ANOVA of the whole design that a statistical analysis plan usually specifies, so the intervals differ from a full-design analysis. When the plan asks for the full model, use other software.

BLQ Handling

Rule	In the app	What it does	When to use it
Rule 1 (default)	Standard: pre-first = 0, post-last = missing	BLQ before the first and between measurable values = 0; after the last = missing	The usual convention for NCA and bioequivalence (see below)

Rule 2	All BLQ → zero	Every BLQ = 0	When the protocol sets all BLQ values to zero
Rule 3	All BLQ → excluded	Every BLQ = missing; the curve then starts from (0, 0) and runs across the removed samples	Sensitivity analyses
Rule 4	Before and between measurable → LLOQ/2, after the last → missing	BLQ before and between measurable values = LLOQ/2; after the last = missing; pre-dose = 0	Sensitivity analyses; needs a justification in the protocol
Rule 5	Before peak → zero, after peak → excluded	BLQ before C _{max} = 0; after C _{max} = missing	When the protocol specifies it
Rule 6	Before first quantifiable → LLOQ/2, rest → zero	BLQ after dosing and before the first measurable value = LLOQ/2; pre-dose, later BLQ and a profile without any measurable value = 0	When the protocol imputes during absorption; it adds area during a lag

Every assay has a lower limit of quantification (LLOQ): the lowest concentration it can measure reliably. Samples below it are reported as BLQ (below the limit of quantification). The app recognises “<0.5” (also “<0,5”), BLQ, BQL, BLOQ, ND (not detected) and NQ (not quantifiable), in any capitals. Set the LLOQ above 0 so the chosen rule is applied to these entries; each rule works per profile, in time order. NS (no sample), N/A and “missing” are treated as missing values, not as BLQ. If you click Process Data with the LLOQ left at 0, the app stops; when the entries look like “<0.5”, it shows the value it found and a button Set LLOQ to 0.5, which fills in the field (Tutorial 2).

ICH M13A (2.2.2.2) treats BLQ values as zero in PK parameter calculations and leaves them out of the half-life. The app never uses a value set by a rule for the half-life or the lag time. Rule 2 sets every BLQ value to zero. Rule 1 does so before and between measurable values but leaves values after the last one missing, which gives the same AUC to last point. With Rules 3 to 6 the Bioequivalence page reminds you that the protocol has to justify the rule.

In the processed data the app adds a BLQ_flag column marking every value the rule set. It appears in the data preview, and the partial AUC notes and the bioequivalence table use it to report how much of a result rests on such values. A column of that name in your file is replaced.

The rule matters most for partial AUCs. An early window often holds BLQ samples, which Rule 1 sets to 0 and Rule 6 to LLOQ/2, and a late window over a plateau near the LLOQ can rest mainly on values the rule produced. The app says so when more than half of the samples in a window come from the BLQ rule, or when fewer than three measurable concentrations are left in it.

Common Mistakes

Wide format: subjects as columns instead of rows. **Time not reset:** in a crossover, time restarts at 0 in every period. **A negative pre-dose time:** set it to 0. **0 for a missing sample:**

leave the cell empty, or write NS. **Duplicate time points:** usually several analytes, matrices or periods in one column; filter the file, or map the Treatment and Period columns. **LLOQ left at 0:** processing is blocked when BLQ text is present. **Dates or clock times as time:** the Time column must be elapsed time since the dose. **Mixed or wrong units:** select the units that match the file; they drive the clearance and volume conversion. **Dose in mg/kg with the unit set to mg:** see Units and Doses. **Subject IDs that restart in each sequence:** give every subject its own ID.

A cutoff with no sample near it: the protocol asks for AUC 0–30 min, but the samples are at 15 and 60 min. The app interpolates the concentration at 30 min, so the value depends on the trapezoidal method.

Excluding a subject by deleting rows: the app has no control to exclude a subject or period, and the Analysis Record then holds only the edited file. Follow the procedure in Exclusions, Outliers and Deviations, in the chapter on regulated use.

Using the App

This chapter describes each path, with the labels the app shows. The Tutorials chapter takes you through them step by step; Reading Your Results explains the tables.

Path 1: Plan a Study

Calculates the sample size, or the power for a given number of subjects, with the PowerTOST package. The page has four cards. **Step 1: What do you want to calculate?** chooses between **How many subjects do I need? (find sample size)** and **What is the chance of success for my planned study? (calculate power)**. **Step 2: What kind of study?** chooses the method: standard average bioequivalence, highly variable drugs (EMA ABEL or FDA RSABE), or narrow therapeutic index drugs (FDA NTID). **Step 3: Study design** offers the designs of Bioequivalence except the paired comparison: 2×2 , $2 \times 2 \times 3$, $2 \times 3 \times 3$, $2 \times 2 \times 4$ and parallel groups for standard bioequivalence; the replicate designs for ABEL and RSABE; only $2 \times 2 \times 4$ for NTID. **Parameters** holds the numbers. Click **Calculate**; a result is cleared when you change an input, so click Calculate again after every change.

The CV the planner asks for depends on the design and the method:

Design and method	CV asked for	Field label
Crossover or replicate, standard bioequivalence	Within-subject CV	Within-subject variability (CV %)
Parallel groups	Total CV (between plus within subjects)	Total CV, between + within subjects (CV %)
ABEL, RSABE, NTID	Within-subject CV of Test and of Reference	Within-subject CV of the Test product (CV %) and of the Reference product (%)

Enter the expected Test/Reference ratio (the default follows the method, as in PowerTOST: 95% for standard bioequivalence, 90% for highly variable drugs, 97.5% for narrow therapeutic index drugs) and, for sample size, the target power. The significance level (0.05) and, for standard bioequivalence, the acceptance limits can be changed under Advanced: acceptance limits & significance level. After a log-transformed bioequivalence analysis of your own data, a button such as Use the Peak Concentration (C_{max}) within-subject CV from my BE analysis (1.8%) offers its CV; with several metrics, Metric from the BE analysis chooses which. It offers the within-subject CV of a crossover for a crossover design, the total CV of a parallel analysis for a parallel design, and CV_{wT} and CV_{wR} of a replicate analysis for the scaled methods. The spread of C_{max} between subjects in batch results is not a within-subject CV, so the app does not offer it. The results are in three tabs: Power vs. Number of Subjects, Sample Size vs. Variability, and Calculation Summary.

For a pivotal partial AUC, plan with the within-subject CV of that partial AUC. An early window is often more variable than total AUC, so a CV taken from AUC gives too few subjects.

The planner makes no record. For a protocol, write down the method, design, CVs, expected ratio, power, significance level, the resulting N, and the versions of the app and PowerTOST (About page), and keep a screenshot of the Calculation Summary.

Path 2: Upload & Check Data

Choose **What kind of file? (Simple table (one row per sample) or CDISC ADNCA dataset)**, upload it, and set the Delimiter and Decimal point for a text file. Check the column mapping (Subject ID, Time, Concentration, and where present Treatment / Formulation, Period, Sequence and Dose), set the **LLOQ value** and the **BLQ handling rule**, and click **Process Data**. The app runs more than 30 quality checks: errors block processing, warnings do not. Some checks stop data the app cannot analyse safely: a CDISC-style file uploaded as a simple table, mixed units, dates or clock times in the Time column, profiles that do not start near time zero (time since the first dose), several profiles stacked in one column (duplicate times), subject IDs that restart in each sequence, and a decimal point in a decimal-comma file. A second dose inside one profile, or a multi-period file without a mapped Period, gives a warning. Processing a new file clears all previous results. Afterwards a panel links to the next steps: Visualize Data, Single Subject, Batch Analysis and Bioequivalence. Reading Your Results explains the quality report.

Path 3: Visualize Data

Creates concentration–time figures from the processed data, from the hub or with the Visualize Data button on the Upload page. The settings are in the sidebar on the left; the plots are stacked on the right, and the **Plot Type** card decides which plots appear.

Plot Types

Plot Type has three options. Individual Profiles (spaghetti plot) draws one concentration–time line per profile (subject, treatment and period), coloured by a grouping of your choice; use it to look at individual variability, outliers and carry-over. **Summary Plot** (geometric mean $\times/\div$ geometric SD) shows the geometric mean at each time point, with error bars from the geometric mean divided and multiplied by the geometric SD. Treatment groups are overlaid automatically. Both plots shows the spaghetti plot above the summary plot.

Concentrations at a time point are usually skewed to the right, so the geometric mean describes the typical profile better than the arithmetic mean. The **Summary statistic** dropdown also offers the arithmetic mean $\pm$ SD, labelled as arithmetic and with a warning. Values of zero, for example BLQ results set to 0, cannot enter a geometric mean and are left out at that time point, so at time points where most samples are BLQ the mean is biased upwards; the figure legend counts the values left out.

Sidebar Controls

Y-axis scale (Linear or Log₁₀ (semi-log)): the semi-log view makes the terminal elimination phase linear, which shows biphasic profiles and half-life problems. Zero concentrations cannot be shown on a log scale; they are omitted with a note.

Colour lines by (spaghetti plot): Subject ID (always available), Treatment, Period and Sequence (when that column was mapped), and None. In a crossover, colour by Period to look for period effects, or by Treatment to compare formulations. **Show data points on lines** adds the samples to the spaghetti plot; **Overlay individual observations** adds them to the summary plot. **X-axis tick spacing** sets the distance between time labels (Auto by default).

Normalise to dose (show C/Dose), a checkbox that appears only when a Dose column was mapped: concentrations are divided by the dose, so the Y axis shows C/Dose (for example ng/mL per mg). A missing or zero dose gives a missing value. Change the Y-axis label to match, for example “C/Dose ($\text{ng}\cdot\text{mL}^{-1}\cdot\text{mg}^{-1}$)”.

Shade the partial AUC intervals of the last analysis shades on the summary plot the intervals of the last batch or bioequivalence run; the Figure Record reproduces the shading.

Colour palette: Default (app palette), Colorblind-safe (Okabe-Ito), or Grayscale. For print, choose Grayscale and export at 600 DPI. **Theme:** Clean (white background), recommended for reports, Minimal, or Classic. The X-axis label, Y-axis label, Plot title and Base font size (pt) (8–18) can be edited. The default Y-axis label is “Concentration”; add the drug and unit, for example “Cyclosporin (ng/mL)”. The default X-axis label is “Time (h)”.

Suggested Figure Legend

Above the plots, a card titled Suggested figure legend holds a ready-to-copy description of the current figure: number of subjects and observations, plot type, summary statistic, error bars, treatment names, the number of concentrations ≤ 0 left out of the geometric mean, **Y-axis scale** and colour grouping. It updates when you change settings. Paste it into your report and add the drug name and units.

Exporting a Figure

Export controls are at the bottom of the sidebar under **Export Figure**, always visible. A small “Will export” label above the button shows which plot will be downloaded; with Both plots, Which plot to export? chooses one. Format: PNG (Word documents, presentations, email), PDF (vector; recommended for journals that accept PDF figures), or SVG (to edit in Inkscape or Adobe Illustrator). Width (in) and Height (in) take 2–20 inches; journals usually ask for 3.0–3.5 in for a single column and 6.5–7.0 in for a double column. Resolution (DPI): 150 (screen and drafts), 300 (print, default) or 600 (high-resolution submission). **Download Figure** is grey until the first plot has appeared. The file name includes the plot type (spaghetti_plot.png or summary_plot.png), so both can be downloaded without overwriting each other.

The plots on screen are interactive (plotly): hover to see subject, time and concentration; drag to zoom; reset with the home icon. The exported file is a separate static figure (ggplot2) built from your sidebar settings.

Path 4: One Subject at a Time

Under **Where is your data?**, choose **Use my uploaded dataset (from step 2)** to step through the profiles with the arrow buttons, or **Enter data manually (type values directly)** to type times and concentrations, one per line, and click **Use This Data** (at least 3 points; Tutorial 1). Under **Analysis Settings**, set the Dose (filled in from the Dose column when there is one), the units, the **Route of administration** and the minimum R^2 , and click **Run PK Analysis**. The **PK Parameters** card shows the results; the **Half-Life Calculation** card shows the terminal-phase points, which you can change under **Points for half-life:** and **Recalculate**. The result belongs to the profile and settings it was computed with: selecting another profile, or changing a setting, clears it. Downloads: **Download Results (CSV)** and a Complete Analysis Record.

Partial AUCs (optional): set **Number of intervals** and enter the start and end of each, in the time unit of your data. Tick **End at the last measurable concentration (t)** for an interval that runs to the end of the profile, and **Also report Cmax and Tmax in this interval** when the protocol asks for the peak inside a window. An interval that reaches past the last measurable concentration stays empty, with a note saying so: partial AUCs are never extrapolated.

Path 5: All Subjects (Batch)

The sidebar has three cards. **Administration:** **Route of administration**, **Dose information (Same dose for all subjects, or Doses differ by subject or period (from Dose column in data))**, the dose, the infusion duration for an infusion, and **Steady-state (drug given repeatedly)** with the Dosing interval τ . **Units:** Dose unit, Time unit, Conc unit, and the molecular weight for molar units. **Calculation Method:** **Trapezoidal method (Linear-up / Log-down**, the default, or **Linear-up / Linear-down)**, **Minimum R² for half-life estimation** (default 0.70), and **Calculate dose-normalised parameters**. Then **Partial AUCs (optional)** and **Run Analysis on All Subjects**.

The results are in five tabs. Concentration-Time Profiles shows all profiles and a Mean $\pm$ SD plot; that mean is arithmetic, so use Visualize Data for geometric summaries. **All Parameters (per profile)** is the table, with a short default view and **Show all parameters** for every column; AUC % Extrapolated above 20% is flagged in amber. **Summary Statistics** gives the statistics per treatment. Individual Profiles (grid view) shows one small plot per profile, and **Half-Life Review** lets you choose the terminal-phase points of any profile: select it under Profile:, tick the points under **Points for half-life:** and click **Recalculate**. A selection stays with its profile; **Remove override (automatic fit)** returns it to the automatic fit. A CDISC parameter codes panel lists the official code of each parameter.

When the automatic terminal fit of a profile has an adjusted R² below the minimum, its half-life, AUC $_{\infty}$, CL/F, Vz/F and MRT are left empty and a note names the profile; select the points in **Half-Life Review** if you want a value. Profiles with fewer than 2 positive concentrations are left out, with an amber note listing them. A single-dose run also notes profiles whose pre-dose concentration is above 5% of Cmax. At steady state, parameters extrapolated to infinity and the single-dose volume and MRT are left empty. Changing a setting clears the results, so run the analysis again.

Partial AUCs join the table, the summary statistics and the downloads. An interval that reaches past the last measurable concentration of a profile stays empty for that profile, and a note names the profile and its last measurable time. With dose normalisation switched on, the partial AUCs and the Cmax within an interval are divided by the dose as well; the time of that maximum is not.

Downloads: **Download Results (CSV)**, **Download Results (Excel)**, with summary statistics per treatment and a sheet of CDISC codes, and a Complete Analysis Record. For a study report, use the record: the plain downloads carry no app version, data hash or settings.

Path 6: Bioequivalence

Runs the NCA, then the ANOVA, then the confidence interval and a forest plot, and gives a verdict where one is justified.

Step 1: PK Analysis Settings

Dose information, dose, units, **Route of administration** (oral, IM or SC, or IV bolus; an IV infusion is not supported here), trapezoidal method and minimum R^2 , as in Path 5, and **Steady-state (drug given repeatedly)** with τ for steady-state data. At steady state AUC Within Dosing Interval replaces AUC to last point and AUC to infinity; after a first run, Minimum Concentration (Cmin) and Concentration at Tau (Ctau) can be added to the parameters to compare.

Step 2: Bioequivalence Settings

Study design: 2×2 standard crossover, 2×2×3 or 2×2×4 full replicate, 2×3×3 partial replicate, parallel groups, or paired comparison. The design found at upload is pre-selected, and a crossover chosen for one-period data is flagged. If all subjects received the treatments in the same order, the data are analysed as a paired comparison.

Reference treatment: the ratio is Test ÷ Reference. The run is blocked until one is chosen, and the results table shows the comparison, for example “New / Old”.

Statistical model: leave it at **Fixed effects (all terms fixed)** unless the protocol specifies the mixed model. The model has Sequence, Subject, Period and Treatment as factors; with fixed effects this is EMA Method A, and the mixed model (random subject intercept) is EMA Method B, not the FDA model for replicate designs. Fixed-Effects vs. Mixed-Effects Models, in Statistical Methods, explains the difference.

Log-transform (recommended), **Parameters to compare** (Cmax and AUC to last point by default, the primary parameters of ICH M13A; AUC to infinity, Tmax and half-life can be added), **Confidence interval** (80 to 99%; a verdict is given only at 90%) and the acceptance limits **Lower (%)** and **Upper (%)**. Click **Run Complete BE Analysis**. Changing a setting clears the results; the parameters you chose are kept.

Widened and Narrowed Limits

With limits wider than 80–125%, two more settings appear. **Widened limits apply to** chooses Cmax only, Cmax and partial AUCs (not AUC to last point or to infinity), or All compared metrics (e.g. no-effect boundaries); the default is Cmax only. Also require the point estimate within 80.00–125.00% is ticked; untick it only for comparisons without such a rule, such as drug-interaction no-effect boundaries.

Widened limits are not ABEL or RSABE. Entering limits wider than 80.00–125.00% does not perform a reference-scaled analysis: the app does not check the conditions for widening (a pre-specified, clinically justified widening, CVwR of this study above 30%, the cap at 50%, a CVwR not driven by outliers; EMA 2010, section 4.1.10), and it does not compute the FDA scaled criterion. Use widened limits only when the protocol fixes them in advance, and document the justification outside the app. For replicate designs a table shows the within-subject variability of the Reference (and of the Test when replicated) and the EMA widened limits it would imply, for information only.

For a narrow therapeutic index drug under the EMA approach, enter 90.00 and 111.11. Narrowed limits apply to every compared metric; the FDA NTID scaled test and its variability comparison are not available.

Running and Reading the Results

The tabs are **Test vs. Reference Comparison** (the table and the forest plot), PK Parameter Table, Statistical Details (ANOVA), Concentration-Time Profiles, Individual Profiles and **Half-Life Review**. Downloads: **Download Complete BE Report (Excel)**, **Download CI Table (CSV)** and a Complete Analysis Record. The verdict uses confidence limits rounded to two decimals. A design summary and an amber warning for subjects missing a treatment appear after the NCA. Reading Your Results explains the columns.

With the results the app lists points that ICH M13A asks you to check. A profile with fewer than 2 measurable concentrations has no NCA result and is counted under Profiles missing; its subject leaves the comparison, which M13A (2.2.1.1) accepts only as a planned exception, in general for no more than one subject. A period whose AUC to last point is below 5% of the geometric mean of that treatment is flagged as very low exposure, on the same terms. A single-dose study also lists a pre-dose concentration above 5% of C_{max}, and AUC to last point covering less than 80% of AUC_∞ in more than 20% of the profiles; neither check runs at steady state. Fewer than 12 evaluable subjects is checked on the smaller of the Test and Reference counts.

A verdict is computed on all the data you uploaded. When the checks name a period with a pre-dose concentration above 5% of C_{max}, ICH M13A (2.2.3.3) excludes it from the primary analysis, so the verdict shown is not the M13A primary analysis: exclude the period as the protocol says (see Exclusions, Outliers and Deviations) and run again. With fewer than 12 evaluable subjects, M13A (2.2.3.1) does not accept the study as pivotal, and the verdict is a statistical result only.

For parallel groups the app also calculates the Welch interval, which does not assume equal variances, and reports it when it would change the conclusion. The pooled interval stays the primary result; if your statistical analysis plan or a guidance asks for an unequal-variance analysis, the app's verdict is not your primary result.

When There Is No Verdict

The Bioequivalent? column says why.

Situation	What the app shows
A crossover without a mapped Period column	No estimate and no verdict; the reason says to map Period
A confidence level other than 90%	The interval, without a verdict
The mixed model was chosen and cannot be fitted	The fixed-effects result for information, no verdict: changing the model after seeing the data is a post-hoc change (ICH M13A)
A paired comparison (a single treatment order)	Ratio and interval, no verdict: period and treatment cannot be separated
Half-life, T _{max} , or a parameter without log transformation	A ratio (half-life) or a difference, no verdict: not bioequivalence endpoints
A supportive partial AUC	Ratio and interval, no verdict
A metric with a value of zero in any profile	No estimate and no verdict, with the zero counts per treatment

Tmax is compared as an untransformed difference with a t-based interval. Tmax is discrete and is usually reported as median and range, with a nonparametric test when the plan asks for one; do that outside the app, and ignore geometric statistics for Tmax.

Partial AUCs in Bioequivalence

Partial AUC intervals here also carry a role, set under **Role in bioequivalence**. A **Pivotal (verdict)** interval receives a bioequivalence verdict, a **Supportive (no verdict)** interval only its ratio and confidence interval. The metrics appear under **Parameters to compare**, ticked, as soon as the intervals are valid, and Cmax within an interval can be compared as well. A metric whose value is zero in any profile receives no estimate and no verdict, because zero cannot be log-transformed and the profiles that would drop out are the low-exposure ones. That metric keeps its row in the table and in the forest plot, with the reason, the zero counts per treatment and the number of profiles that have no value.

Analysing Drug–Drug Interaction Studies

Bioequivalence (Path 6) can also analyse drug–drug interaction (DDI) studies: the design compares two conditions within subjects, with an ANOVA model and a confidence interval for the geometric mean ratio.

How to Set Up a DDI Analysis

Prepare the data as a crossover dataset (Scenario 4 or 5). Choose the victim drug given alone as the **Reference treatment**, so the ratio is (with perpetrator) ÷ (alone): the fold-change in exposure. For a single-dose victim, tick AUC to infinity under **Parameters to compare** as well; at steady state compare AUC Within Dosing Interval, Cmax and Cmin.

Run the analysis. A ratio of 200% means the perpetrator doubled the victim's exposure; 50% means it halved it. Check the Comparison column to confirm which treatment is in the denominator.

Interpreting DDI Results

A DDI study quantifies how large the interaction is; it is not a test of equivalence. ICH M12 (section 5.2) asks for the geometric mean ratio with and without the precipitant and its 90% confidence interval for AUC and Cmax (and Cmin when relevant), and for the range of individual ratios. The app does not give individual ratios: compute them from the exported PK Parameter Table. Interpret the result against no-effect boundaries for the victim drug, preferably based on its exposure-response relationship (section 5.3.1). The default no-effect boundaries are a 90% CI within 80–125%, which M12 calls generally acceptable but overly conservative for most drugs.

If the study used a sensitive index substrate, the point estimate also classifies the precipitant (M12 section 5.3.2): a strong, moderate or weak inhibitor raises the substrate's AUC at least 5-fold, 2- to 5-fold, or 1.25- to 2-fold; a strong, moderate or weak inducer lowers it by at least 80%, by 50 to 80%, or by 20 to 50%. These classes describe the precipitant, not the size of the interaction for other victim drugs.

Practical Considerations

To compare with no-effect boundaries, enter them as acceptance limits, choose All compared metrics under **Widened limits apply to**, and untick the point-estimate

requirement if it does not apply. A YES in the Bioequivalent? column then means that the 90% CI lies within the boundaries you entered; it is not a bioequivalence conclusion.

For fixed-order DDI studies (all subjects receive the victim alone first, then with the perpetrator), select Paired comparison. The app reports the ratio and its confidence interval without a verdict, because period and interaction effects cannot be separated: read the interval against the boundaries yourself.

The forest plot and confidence interval table can go into the DDI section of a clinical study report when they were produced on a qualified installation and reviewed according to your procedures.

Complete Analysis Record

Available from One Subject at a Time, All Subjects and Bioequivalence, with **Generate Analysis Record**; Visualize Data produces a similar Figure Record. Fill in **Analyst name (optional)** and **Study name (optional)** first: they appear in the record, but they are typed, not signed in, so they do not establish who did the analysis.

The zip contains results.xlsx, app_results_reference.csv, analysis_settings.json, nca_pipeline.R (the app's own data-processing code), reproduce_analysis.R, reproduction_check.txt, data_integrity.txt, analysis_summary.html and a copy of the data file. From the batch and bioequivalence paths, results.xlsx has a Checks sheet, and the summary a Checks and Notes section, with the data quality findings and the notes shown with the results, such as the ICH M13A checks. data_integrity.txt holds SHA-256 hashes of the data, the settings, the results, the reference results, the script and the pipeline code. A SHA-256 hash is a 64-character fingerprint that changes if a single character in the file changes.

When it creates the record, the app runs the reproduction script in a separate R process and writes the outcome to reproduction_check.txt:

Outcome	Meaning
MATCH	The shipped pipeline, run again on the enclosed data and settings, gives the same NCA parameters (largest relative difference below 10^{-6})
CLOSE	The same within a relative difference of 10^{-3} , for example after a package update that changes rounding
DIFFERENT	Different results, or a data file, pipeline code or reference results that were changed, or a parameter present on one side only
NOT COMPARED	The reference results (app_results_reference.csv) are missing

What the record shows and what it does not. MATCH shows that the analysis can be reproduced with the same code and packages. It is not independent verification. For a bioequivalence record the script recomputes the NCA parameters only: the ANOVA, the confidence intervals and the verdict are recorded in results.xlsx and are not recomputed, and the script does not check results.xlsx. The manifest is not signed: whoever changes a file

can also recompute its hash. To make the record tamper-evident, store the zip, or its SHA-256, in a controlled system. The record is not an audit trail in the sense of 21 CFR Part 11 or EU GMP Annex 11; it documents one analysis. Timestamps come from the clock of the computer that ran the app (UTC on shinyapps.io), and the summary names the time zone.

If the uploaded file was no longer available when the record was made, the record holds the table as the app read it, written as CSV, with a different name and hash; the settings and the Checks say so. Keep the original source file with the record.

Partial AUC intervals, their roles and the BLQ share used for the notes are recorded, and the reproduction script recalculates the partial AUCs. Change an interval in the settings file afterwards and the check reports DIFFERENT.

Re-running a record. Unzip it into an empty folder. In R, set the working directory to that folder (`setwd("path/to/folder")`) and run `source("reproduce_analysis.R")`. The script checks the data hash, recomputes the results and prints the outcome.

Reading Your Results

The Data Quality Report

After Process Data, the Upload page lists what the checks found, one line per finding.

Column	Meaning
Severity	ERROR blocks processing; WARNING and INFO do not; OK means the check passed
Category	What was checked: Structure, Time, Concentration, Subjects, Design, Format and others
Message	What the check found, for example "22 BLQ/text entries detected"
Detail	The rows, subjects or values concerned
Action	What to do about it, where there is something to do

Fix every ERROR in the file or the mapping, and process the data again. Read the warnings too. They do not stop the analysis, but they can change what the results mean.

The Parameter Table

All Parameters (per profile) in Path 5, and PK Parameter Table in Path 6, have one row per profile: a subject under one treatment in one period. Subject, Treatment and Period identify the row; the other columns are parameters, labelled in words with their unit, for example "AUC to Last Point (ng/mL·h)". The default view shows the main parameters; Show all parameters shows every column. Appendix A lists each parameter with its definition, unit and when it is reported. An empty cell is not an error: it means the parameter could not be estimated for that profile, and a note above the table says why (for example a terminal fit below the minimum adjusted R^2).

Summary Statistics

Summary Statistics gives, per treatment and parameter:

Column	Meaning
N	Number of profiles with a value
Geometric Mean	$\exp(\text{mean of } \ln \text{ values})$; the usual summary for AUC and Cmax
Geometric CV (%)	$100 \cdot \sqrt{(\exp(\text{SD}^2 \text{ of } \ln \text{ values}) - 1)}$
Median, Q1 (25%), Q3 (75%)	The middle value and the quartiles; the usual summary for Tmax
Arithmetic Mean, Std Dev, CV (%)	Mean, standard deviation, and SD as a percentage of the mean
Min, Max	The range

Geometric statistics need positive values; they are not meaningful for Tmax or for a parameter that can be zero.

The Bioequivalence Table and Forest Plot

The table on the Test vs. Reference Comparison tab has one row per compared metric.

Column	Meaning
PK Parameter, Comparison	The metric, and which treatment is divided by which ("Test / Reference")
Subjects (Test), Subjects (Reference)	Subjects that enter the comparison
Estimate	The geometric mean ratio Test ÷ Reference in percent (a difference for Tmax or untransformed parameters)
90% CI Lower, 90% CI Upper	The confidence interval, at the level you chose
Accept. Lower, Accept. Upper	The acceptance limits used for this metric
PE within 80–125%	Whether the point estimate lies within 80.00–125.00%, for widened limits
Bioequivalent?	YES or NO, or the reason why there is no verdict
Profiles missing, Zero values, Mostly BLQ (Test and Reference)	Profiles without a value, with a value of zero, or resting mainly on values set by the BLQ rule
Model	The model fitted, and whether a mixed model failed

The forest plot shows each estimate as a point with its confidence interval as a horizontal bar, and the acceptance limits as vertical lines: a bar entirely between the lines is a YES. The Residual Variance (MSE) and Degrees of Freedom are in the downloads (Download CI Table (CSV)); the within-subject CV of a crossover is $100 \cdot \sqrt{(\exp(\text{MSE}) - 1)}$.

Statistical Details (ANOVA) shows, per metric, the model terms (Sequence, Subject, Period, Treatment) with their degrees of freedom and F tests. A significant Sequence or Period term is a reason to look at the data, not a reason to change the analysis.

Judging the Terminal Phase

Half-life, AUC to infinity, CL/F, Vz/F and MRT all rest on the terminal slope λ_z . Before you report them, check each profile against these points, and follow your statistical analysis plan where it is stricter.

- **At least 3 points** in the terminal log-linear phase (ICH M13A, 2.1.8), after Tmax and not counting Cmax itself. The app accepts 2 points in a manual selection, but a 2-point fit has no measure of fit and should not be reported.
- **Span:** as a rule of thumb the points should cover at least two half-lives; ICH M13A (2.1.8) asks for sampling over at least three.
- **Adjusted R²:** the app's default minimum of 0.70 is a floor, not a recommendation; many analysis plans use 0.80 or 0.90.

- **AUC % Extrapolated** above 20% (amber in the table): the extrapolated part carries much of AUC to infinity. Report AUC to infinity, CL/F, Vz/F and half-life with a flag, or leave them out of the descriptive statistics, as the plan says.

Half-Life Review shows the fit the results use, on a log scale, with the chosen points marked.

Checking Plausibility

- **Pre-dose concentration above zero:** carry-over, earlier dosing or an endogenous compound. In a bioequivalence study a value above 5% of C_{max} takes the period out of the primary analysis (the app says so).
- **C_{max} at the first sample after the dose:** absorption was faster than the sampling, and C_{max} may be underestimated. ICH M13A (2.1.8.1) asks for this to be discussed.
- **T_{max} at the last sample:** absorption was still going on, or sampling stopped too early; there is then no terminal phase and no λ_z .
- **A half-life much longer than expected:** check the number of points and the % extrapolated (Tutorial 3).
- **Flip-flop kinetics:** with slow absorption, such as a modified-release product, the terminal slope reflects absorption rather than elimination. Point selection cannot correct this: report the value as an apparent half-life. Only IV data can tell the two apart.
- **A profile without measurable concentrations** (non-absorber): it has no NCA result, and in bioequivalence it is counted as missing and named in the checks.
- **Very different results under two BLQ rules:** the result depends on the imputation; report the rule and consider a sensitivity analysis.

Using NCA Assistant in a Regulated Environment

This chapter is for users who report results from the app in a clinical study, a regulatory submission or another setting under a quality system. It does not replace your organisation's procedures for computerised systems.

Status and Scope

NCA Assistant is open-source software without warranty or supplier support, and it is not validated for your use. It performs average bioequivalence only, not ABEL or RSABE. The public instance on shinyapps.io is outside the scope of any qualification: its package versions are set by the deployment, not by the validated release.

Which Rules Apply

Bioequivalence and drug interaction studies are clinical trials, so good clinical practice applies: ICH E6(R3) and the EMA guideline on computerised systems and electronic data in clinical trials (EMA/INS/GCP/112288/2023). EU GMP Annex 11 and 21 CFR Part 11 apply where your organisation or the use falls under them. Your own procedure for validating computerised systems decides what is needed; the validation package gives the evidence to start from.

Installing a Qualified Instance

Download a tagged release (github.com/Robterheine/NCAassistant/releases), install it on a computer or server whose access you control, and install the validated package versions with `Rscript install_and_run.R --validated` (see [Installing the App on Your Own Computer](#)). The plain `install_and_run.R` installs the latest packages from CRAN, which is fine for evaluation but is not the qualified configuration.

Qualification

The validation folder holds the User Requirement Specification, the IQ/OQ/PQ protocol and the test script. In outline:

1. Review the URS and its risk assessment against your intended use, and approve the protocol before execution (protocol section 1.2).
2. Run `Rscript validation/validation.R` and keep `validation_results.csv` and `validation_environment.txt`. The installation checks IQ-REL-01 and IQ-REL-02 compare your files and package versions with the release.
3. Carry out the manual OQ tests (protocol section 4) and the user PQ (section 5): analyse datasets like your own studies and compare the results with an independent reference.
4. Assess every deviation and every SUPPORTIVE failure, and have QA sign the outcome (sections 7 and 8).

The checklist for adopting organisations in section 1.3 of the protocol lists these steps.

Controls the App Does Not Provide

The app has no user accounts, no audit trail and no electronic signatures. Put these controls in place around it:

- access to the qualified installation restricted to trained users;
- a log entry for every analysis: who ran it, when, on which data file (with its SHA-256), with which release, and for what purpose;
- a second person who checks the settings against the protocol and the results before they are used;
- archiving of the source files and the Analysis Records in a system with an audit trail.

Before Each Analysis

Check on the About page that the version and the pipeline SHA-256 are those of your qualified release (the release manifest, `validation/release_manifest.csv`, lists the hash of every file). Record the SHA-256 of the source data file. Check the settings against the protocol and the statistical analysis plan: design, Reference treatment, model, acceptance limits and their scope, BLQ rule and LLOQ, partial AUC intervals and their roles.

Exclusions, Outliers and Deviations

The app has no control to exclude a subject or a period, so an exclusion means preparing a second data file. ICH M13A (2.2.1) asks for the criteria to be defined in the protocol and for any exclusion to be documented before the bioanalysis. Extreme values should not be removed on statistical grounds alone; only contemporaneously documented protocol violations justify removal (M13A 2.2.1.1). A subject without measurable concentrations, or with very low ones (AUC below 5% of the geometric mean), may be excluded only as an exception, in general for no more than one subject. A period with a pre-dose concentration above 5% of C_{max} is excluded from the primary analysis (2.2.3.3).

A procedure that keeps the chain from source to result intact:

1. Record the SHA-256 of the unedited source file.
2. Make the analysis file with a documented, repeatable step, for example a short R script under version control, rather than by hand.
3. Keep both files, the list of exclusions with the reason for each, and the protocol reference.
4. Run the analysis on the analysis file and create the Analysis Record.
5. File the source hash and the exclusion list with the record.
6. Where the plan asks for it, run the analysis with and without the excluded data and report both.

Reading Verdicts in a Regulated Context

- A verdict is computed on all the data you uploaded. When the checks name data that ICH M13A excludes, the verdict is not the M13A primary analysis (Path 6, Running and Reading the Results).

- Widened limits are not ABEL or RSABE, and All compared metrics also widens AUC to last point and AUC to infinity, which no guideline allows for bioequivalence. To widen a partial AUC, choose C_{max} and partial AUCs.
- ICH M13A allows a fixed or a mixed model, but its primary analysis includes only subjects with data for both treatments (2.2.3.2). The mixed model also uses subjects with one treatment, which is not that population; use it only when the plan specifies it.
- For parallel groups the pooled-variance interval is the app's primary result; if the plan specifies an unequal-variance (Welch) analysis, that is your primary result.

The Analysis Record in a Regulated Context

The record documents one analysis and shows that it can be reproduced with the same code and packages (Complete Analysis Record, in the chapter on using the app). It does not re-derive the bioequivalence statistics, it is not signed, and whoever changes a file can recompute its hash. Store the zip, or its SHA-256, in a controlled system. The Analyst name and Study name fields are typed, not signed in: record who ran the analysis in your log. For a manual selection of half-life points, the record keeps the final selection but not the reason: write the reason in the study file.

Reporting and Citing

Cite the software with its version: ter Heine R. NCA Assistant (v1.7.0). Radboud Applied Pharmacometrics, Radboudumc, Nijmegen, The Netherlands.
<https://github.com/Robterheine/NCAassistant>. Add the pipeline SHA-256, the reference of your qualification, and the SHA-256 of the Analysis Record. The Example Methods Paragraphs on the Statistical Methods page describe the calculations; they say the analysis follows ICH M13A, EMA and FDA guidance, so check that your study design and analysis do before you keep that sentence. For a sample-size justification, record the planner's inputs and result (Path 1).

Archiving and Retention

File the source data file, the analysis file, the Analysis Record, the SHA-256 of the zip, the release tag, validation/renv.lock and the qualification reference together. Keep them as long as your procedures and the law require; in the EU the trial master file of a clinical trial is kept for at least 25 years (Regulation (EU) No 536/2014, Article 58). The app keeps nothing: download the record before you close the session.

Upgrades and Change Control

Read the version history on the About page before you upgrade. Entries marked "Results can differ" and "Correctness fix" tell you which analyses a new version may change. A new release, a new R version or a new package version needs at least a targeted re-qualification (URS section 7); decide in advance whether completed analyses are repeated.

Data Protection

Use a local installation for real study data. Pseudonymised data are still personal data under the GDPR (Article 4(5)), and pharmacokinetic data from a trial are health data (Article 9).

Sending them to a hosted service needs a processor agreement and, for a host outside the EU, a transfer basis. Upload only synthetic, example or anonymised data to the public instance.

Reporting Problems

Report problems at github.com/Robterheine/NCAassistant/issues, with a description of the data (never the data themselves, unless synthetic). Correctness fixes are listed in the version history under "Correctness fix", with the situations they affect.

Introduction to Non-Compartmental Analysis

What Is Non-Compartmental Analysis?

Non-compartmental analysis (NCA) is a method for describing the pharmacokinetics of a drug without assuming a specific structural model for how the drug distributes in the body. Instead of fitting a mathematical model, NCA calculates pharmacokinetic parameters directly from the observed concentration–time data using numerical integration and regression.

NCA is the standard approach for pharmacokinetic data in drug development. EMA, FDA and PMDA use it for bioequivalence studies, and it is routine in Phase I trials and dose-finding studies.

The Concentration–Time Curve

After a drug is administered, blood samples are collected at specified times and the drug concentration is measured. When plotted against time, the curve shows absorption (concentrations rising), distribution (transition), and elimination (concentrations declining). NCA extracts quantitative parameters from this curve without assuming its shape.

Key Pharmacokinetic Parameters

Parameter	Symbol	What it tells you
Peak concentration	C _{max}	The highest concentration observed. Relevant for efficacy and safety.
Time of peak	T _{max}	When the peak occurs. Reflects the rate of absorption.
AUC to last point	AUC _{0–t}	Exposure up to the last measurable concentration, from the observed data only. The primary AUC in bioequivalence.
AUC to infinity	AUC _{0–∞}	AUC _{0–t} plus the extrapolated tail, C _{last} /λ _z . Total exposure after a single dose.
AUC % extrapolated	%AUC _{extrap}	The share of AUC _{0–∞} that is extrapolated. Above 20% the value rests largely on the extrapolation.
Elimination rate constant	λ _z	The slope of the terminal phase on a log scale, from a regression on its last points.
Adjusted R ²	R ² _{adj}	How well the terminal regression fits; the app leaves λ _z -based parameters empty below a minimum.
Terminal half-life	t _{1/2}	ln 2/λ _z : the time for the concentration to halve during the terminal phase.

Apparent clearance	CL/F	Clearance divided by the bioavailability F: the volume of plasma cleared of drug per unit time.
Apparent volume of distribution	V _z /F	The terminal volume of distribution divided by F.
Mean residence time	MRT	The average time a drug molecule stays in the body; after an oral dose it includes the absorption time.
Partial AUC	AUC t ₁ –t ₂	Exposure over part of the profile, for example the first 30 minutes. It catches a difference in the shape of the curve that C _{max} and total AUC can miss.

Appendix A lists every parameter the app reports, with its unit.

Why NCA Is Used So Widely

NCA does not assume a compartmental structure, so the same calculations apply whether a drug follows one-compartment kinetics or multi-phasic elimination. Results depend on the observed data and the numerical method, not on model selection, which makes them reproducible across software and analysts.

Regulatory agencies accept NCA because the methods are well-defined and transparent. NCA is the standard approach for bioequivalence with pharmacokinetic endpoints, where C_{max} (rate) and AUC (extent) are the primary endpoints.

When to Use NCA vs. Compartmental Modelling

Use NCA when you need standard PK parameters from observed data: first-in-human studies, bioequivalence, drug interaction studies, dose proportionality, food effect studies. Use compartmental modelling when you need to simulate unobserved scenarios, when you have sparse sampling or irregular dosing, or when you need to understand mechanisms. The two approaches are complementary.

Theoretical Background

Trapezoidal Methods for AUC Calculation

The Linear Trapezoidal Rule

Between consecutive observations at t_i and t_{i+1} with concentrations C_i and C_{i+1} , the area is approximated as a trapezoid:

$$\text{AUC}(i) = (t_{i+1} - t_i) \times (C_i + C_{i+1}) / 2$$

This assumes linear concentration change. During elimination, drug concentrations typically decline exponentially, and the linear assumption leads to systematic overestimation: a straight line between two points on an exponential decay lies above the true curve.

The Logarithmic Trapezoidal Rule

If the concentration declines mono-exponentially between two points, the exact integral yields:

$$\text{AUC}(i) = (t_{i+1} - t_i) \times (C_i - C_{i+1}) / (\ln C_i - \ln C_{i+1})$$

This gives a more accurate estimate during the terminal phase. It requires both concentrations to be positive.

The Linear-Up / Log-Down Hybrid

Proposed by Yeh and Kwan (1978): the linear formula while concentrations rise and the logarithmic formula while they fall. It is the default in NCA Assistant. Choose the method in the protocol: ICH M13A (2.2.2.2) asks for it to be reported, and the two methods give slightly different results (in Tutorial 4, an AUC to last point ratio of 111.25% with linear-up/log-down and 111.16% with the linear method).

Two Special Cases

A fall to zero. The logarithmic formula needs two positive concentrations, so it cannot be used when a BLQ sample set to 0 follows a measurable one. NonCompart then adds no area for that segment. The app uses the linear formula there instead, so the fall to zero is counted in AUC to last point, AUC_∞ and everything derived from them.

An IV bolus with a sample at time 0. After an IV bolus the concentration at time 0 is C_0 , the value just after the injection, not a pre-dose value. The app sets aside samples at or before time 0 and back-extrapolates C_0 log-linearly from the first two concentrations when they decline; otherwise C_0 is the first measured concentration.

Terminal Phase Characterisation and Half-Life Estimation

What Is the Terminal Phase?

After distribution is complete, the concentration declines as $C(t) = A \cdot e^{-\lambda_z t}$, where A is the intercept of the terminal phase. On a log-linear plot, this appears as a straight line with slope $-\lambda_z$. The half-life is $t_{1/2} = \ln(2) / \lambda_z$.

Automated Point Selection

NCA Assistant uses NonCompart's slope selection, the same for the results table and the Half-Life Review. It fits log-linear regressions to the last 3, 4, 5, ... points (after extravascular dosing only points after Cmax are used, and never values set by a BLQ rule) and takes the fit with the highest adjusted R²; fits within 0.0001 of the best prefer more points. If that fit has an adjusted R² below the minimum (slider, default 0.70), the profile gets no half-life, and AUC_∞, CL/F, Vz/F and MRT are left empty too; a note names the profile. This happens when the terminal phase is poorly sampled, when late samples are near the LLOQ, or when elimination is multi-phasic. A lower minimum accepts noisier fits. Judging the Terminal Phase, in Reading Your Results, lists what to check before you report a half-life.

Manual Point Selection

Adjust the points yourself for multi-phasic elimination, enterohepatic recirculation or late samples near the LLOQ. In Half-Life Review, select a profile, tick the terminal-phase points and click Recalculate. The slope must be negative, and although 2 points are accepted, a fit to be reported needs at least 3. NonCompart then recomputes half-life, AUC_∞, CL/F, Vz/F and MRT from those points, with its unit conversions. A hand-picked selection is used whatever its adjusted R², so judge the fit on the plot. Selections are kept when you run the analysis again and are listed in the Analysis Record.

With flip-flop kinetics (absorption slower than elimination, as with many modified-release products) the terminal slope reflects absorption. No choice of points corrects that: report the value as an apparent half-life, and do not derive elimination from it. Only IV data can tell the two apart.

Derived Parameters

With λ_z and the observed last concentration Clast, NCA Assistant (through NonCompart) calculates:

$AUC_{0-\infty} = AUC_{0-t} + Clast / \lambda_z$ (observed); with the predicted last concentration from the regression, the "predicted" variants

$AUC \% \text{ extrapolated} = (AUC_{0-\infty} - AUC_{0-t}) / AUC_{0-\infty} \times 100$

$CL/F = Dose / AUC_{0-\infty}$ and $Vz/F = Dose / (\lambda_z \times AUC_{0-\infty})$, without /F after an IV dose

$MRT = AUMC_{0-\infty} / AUC_{0-\infty}$; after an infusion, minus half the infusion duration

The lag time Tlag is the last sampling time before the first measured concentration. After an IV bolus, AUC % back-extrapolated is the share of AUC between time 0 and the first sample, from the back-extrapolated C0. The Statistical Methods page in the app gives the same formulas.

BLQ Handling: Pharmacological Rationale

Every assay has a lower limit of quantification (LLOQ). How BLQ samples are handled changes AUC. Setting early BLQ samples to 0 is usual before absorption; treating late BLQ samples as missing avoids adding area that was never measured. LLOQ/2 is a compromise that adds a little area wherever it is used. Rule 1, the default, follows the usual convention and ICH M13A, which treats BLQ values as zero. Rules 4 and 6 put LLOQ/2 into the absorption phase: that adds area during a lag, though the lag time itself is not changed, because a value set by a rule is not a measured concentration. They belong in sensitivity analyses, or in a protocol that asks for them. Running two rules and comparing the results

shows how much the choice matters. Beal (2001) compares methods for BLQ data in model-based (population) analysis, not in NCA.

Steady-State Considerations

At steady state AUC_{τ} replaces AUC_{∞} . You enter the dosing interval τ ; AUC_{τ} is the AUC from 0 to τ , interpolated between samples and extrapolated with λ_z when the last sample is missing or earlier than τ (not reported if that fit was rejected). $C_{avg} = AUC_{\tau}/\tau$, fluctuation = $(C_{max} - C_{min})/C_{avg} \times 100$ and swing = $(C_{max} - C_{min})/C_{min}$. C_{min} is the lowest observed concentration from 0 to τ , the pre-dose sample included, and C_{τ} is the observed concentration at τ ; the two differ when the minimum falls after the dose. CL/F and V_z/F are calculated from AUC_{τ} . AUC_{∞} , its % extrapolated, $AUMC_{\infty}$, and the single-dose V_{ss} and MRT are not meaningful during repeated dosing and are left empty. A half-life from one interval is unreliable when the interval does not cover the terminal phase.

Partial AUCs: Interpolation and the End of the Interval

A partial AUC is the area over part of the profile. With both cutoffs on sampling times it is simply the sum of the trapezoids between them.

A cutoff between two samples has to be interpolated: linearly, or log-linearly on a falling segment when you chose linear-up/log-down. The two methods differ most where the curve bends, which is why a protocol that asks for a partial AUC should schedule a sample at the cutoff.

An interval can end at t , the time of the last measurable concentration. That time belongs to the profile, not to the sampling schedule: a later sample that came back below the limit of quantification does not extend it. The app calculates such an interval as AUC_{0-t} minus the AUC from 0 to the start, so an interval ending exactly at the last measurable time gives the same number.

Past that point nothing is reported. Extrapolating with λ_z would turn an observed exposure metric into a model-based one, and it would do so for the profiles that carry the least information.

Statistical Methods

The Two One-Sided Tests (TOST) Procedure

Conventional hypothesis testing ($H_0: \mu_T = \mu_R$) can only reject equality, not prove similarity. Schuirmann (1987) reformulated the problem as two one-sided tests:

$H_{01}: \mu(T)/\mu(R) \leq 0.80$ (ratio too low)

$H_{02}: \mu(T)/\mu(R) \geq 1.25$ (ratio too high)

Both rejected at $\alpha = 0.05 \rightarrow$ ratio lies within $[0.80, 1.25] \rightarrow$ bioequivalent. This is equivalent to: the 90% CI for the ratio falls entirely within $[80\%, 125\%]$. The limits are symmetric on the log scale: $\ln(0.80) = -0.2231$, $\ln(1.25) = +0.2231$.

The ANOVA Model for Crossover Designs

The standard model for a $2 \times 2 \times 2$ crossover:

$$\ln(Y) = \mu + \text{SEQ} + \text{SUBJ}(\text{SEQ}) + \text{PER} + \text{TRT} + \varepsilon$$

SEQ accounts for differences between sequence groups, SUBJ removes between-subject variability from the residual, PER accounts for changes between periods, and TRT is the effect of interest: its coefficient estimates the log-ratio of geometric means. NCA Assistant fits this model for all crossover designs, with Sequence, Subject, Period and Treatment as factors; with fixed effects this is EMA Method A.

Fixed-Effects vs. Mixed-Effects Models

Fixed effects (lm): all terms, including Subject, are fixed (EMA Method A). Subjects without both treatments do not contribute to the comparison. Residual df = $N - 2$ for a balanced $2 \times 2 \times 2$.

Mixed effects (nlme::lme): Subject is a random intercept, and denominator df use the containment method. Subjects with only one treatment still contribute. ICH M13A (2.2.3.2) allows a general linear or a mixed model, but its primary analysis includes all data of the subjects with evaluable data for both products; with the mixed model the app also uses subjects with one treatment, which is not that population. Use the model your protocol or statistical analysis plan specifies, for example the mixed model as a sensitivity analysis. If the mixed model was chosen and cannot be fitted, the app shows the fixed-effects result for information, says so in the Model column, and gives no verdict: the model is not changed after the data are seen. This is EMA Method B (one within-subject variance; results agree with replicateBE). It is not the FDA model for replicate designs, which has separate Test and Reference variances, a subject-by-formulation interaction and Satterthwaite degrees of freedom.

Confidence Interval Construction

$$90\% \text{ CI} = \exp(\delta \pm t(0.95, df) \times \text{SE}(\delta)) \times 100\%$$

where δ is the treatment coefficient on the log scale, SE its standard error from the model, and df the denominator degrees of freedom. The limits are rounded to two decimals before comparison with the acceptance limits, as the EMA guideline (2010, section 4.1.8) and the

FDA (2026) ask. Bioequivalence is concluded when the rounded interval lies within the limits (80.00–125.00% by default) and, for wider limits, the point estimate lies within 80.00–125.00%. A verdict is given only for the 90% interval, which corresponds to the two one-sided tests at $\alpha = 0.05$; at another level the app shows the interval without a verdict. Widened limits apply to Cmax only unless you choose otherwise (Path 6, Widened and Narrowed Limits).

Design-Specific Models

Paired comparison: $\ln(Y) = \mu + \text{SUBJ} + \text{TRT} + \varepsilon$, equivalent to a paired t-test; $df = N - 1$. Ratio and interval only, no verdict.

Parallel design: $\ln(Y) = \mu + \text{TRT} + \varepsilon$. $df = N_T + N_R - 2$. Differences between subjects enter the comparison, so more subjects are needed than in a crossover. The interval assumes equal variances in both groups (pooled variance). The app also calculates the Welch interval, which allows unequal variances, and reports it as a sensitivity result when it leads to a different conclusion. When the analysis plan specifies an unequal-variance analysis, that is the primary result, not the app's verdict.

Replicate designs: the same model, with Period at 3 or 4 levels; each administration enters once. The confidence interval uses average bioequivalence limits. The within-subject variability table uses a separate model per treatment (Sequence, Subject and Period, fitted to subjects with that treatment at least twice), as in the replicateBE package; the app's results agree with replicateBE on its 30 reference data sets.

Tmax is compared as an untransformed difference with a t-based interval, without a verdict. Because Tmax takes only the sampling times, it is usually summarised as median and range and compared nonparametrically, outside the app.

Drug–Drug Interaction Analysis

DDI studies use the same statistics as BE: the ANOVA model, the geometric mean ratio and its 90% confidence interval. The difference is the interpretation: the ratio is the fold-change in exposure caused by the interacting drug, judged against no-effect boundaries for the victim drug (see Analysing Drug–Drug Interaction Studies) rather than BE acceptance limits.

ICH M12 on drug interaction studies (2024), published as guidance by EMA and FDA, asks for the geometric mean ratio of AUC and Cmax (and Cmin when relevant) with and without the precipitant, with its 90% confidence interval (section 5.2). Crossover studies, including one-sequence studies, are preferred over parallel designs (section 3.2.1).

Power and Sample Size for Bioequivalence

Exact power via Owen's Q function for standard ABE. For ABEL, RSABE and NTID, power is simulated in PowerTOST: 100,000 simulated studies for the sample size, 10,000 for power and the power curve. Inputs: within-subject CV (for scaled methods, of Test and Reference), expected ratio, target power and design. A higher CV or a ratio further from 100% needs more subjects; a crossover needs fewer subjects than a parallel design.

Practical guidance: without data, a CV of 30% is a cautious assumption. The expected ratio defaults to 95% for standard bioequivalence, 90% for highly variable drugs and 97.5% for

narrow therapeutic index drugs; target power 80%, or 90% for studies where failure is costly. Check the Sample Size vs. Variability tab to see how N changes with the CV.

Partial AUCs in Bioequivalence

Regulators ask for a partial AUC when two products can share C_{max} and total AUC while the shape of the curve differs. FDA product-specific guidances name the metric and the interval per product; Hopefl et al. (2025) list the guidances published between 2020 and 2024.

The statistics do not change. A partial AUC is log-transformed and analysed with the same model and confidence interval as C_{max} and AUC. The interval comes from the protocol; choosing it after seeing the data inflates the chance of a false conclusion of equivalence.

A pivotal metric has to meet the acceptance limits, a supportive one is reported without a verdict. With several pivotal metrics each has to pass on its own. That is an intersection-union test, so no correction for multiple testing applies.

Zeros need care. An early interval with only BLQ samples can be exactly 0, which has no logarithm. Leaving those profiles out would remove the lowest exposures and push the ratio up, so the app reports no estimate and no verdict for that metric and shows how many zero values each treatment had.

For every metric the table also counts, per treatment, the profiles that had no value, the profiles with a value of zero, and the profiles resting mainly on concentrations set by the BLQ rule. The last count matters because such a profile can pull a ratio on the log scale without ever being zero: under Rule 1 an early window may be exactly zero and get no verdict, while under Rule 6 the same sample becomes LLOQ/2 and the metric keeps a verdict that one imputed value helped to decide.

The EMA allows the acceptance criteria for a partial AUC to be widened in the same way as for C_{max} of a highly variable product (EMA/CHMP/EWP/280/96 Rev1, section 6.8.2.2). The app does not scale limits: for a replicate design it shows CV_{wR} and the limits it would imply, and it applies the limits you enter. To widen C_{max} and the partial AUCs but not AUC to last point or to infinity, choose C_{max} and partial AUCs under Widened limits apply to. The same guideline recommends an early and a terminal partial AUC, split at a pre-specified cut-off such as half the dosing interval, when a single-dose study replaces a multiple-dose study.

Plan such a study with the CV of the partial AUC itself. For loxapine inhalation powder the within-subject CV was above 30% for AUC 0–10 min but below 30% for AUC 0–30 min, which is why the 30-minute window became the pivotal metric (Hopefl et al., 2025).

Regulatory Guidelines

Which guideline applies. For immediate-release solid oral dosage forms, ICH M13A (in effect in the EU since January 2025) is the bioequivalence guideline. The EMA guideline of 2010 still applies to other dosage forms and to the widening of limits for highly variable drugs, which M13A does not cover; later parts of the ICH M13 series address such topics. In the USA, FDA guidances and product-specific guidances apply alongside M13A. Check the current status of each document before you cite it.

ICH M13A Bioequivalence for Immediate-Release Solid Oral Dosage Forms (Step 4, 23 July 2024; in effect in the EU from 25 January 2025; published by ICH, EMA and FDA)

EMA Guideline on the Investigation of Bioequivalence (CPMP/EWP/QWP/1401/98 Rev. 1, 2010): <https://www.ema.europa.eu/en/investigation-bioequivalence-scientific-guideline>

EMA product-specific bioequivalence guidance, and the EMA questions and answers on bioequivalence, which describe Methods A and B: on the EMA website

FDA Statistical Approaches to Establishing Bioequivalence (May 2026; replaces the 2001 guidance): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/statistical-approaches-establishing-bioequivalence>

FDA Bioequivalence Studies With Pharmacokinetic Endpoints for Drugs Submitted Under an ANDA, and **FDA product-specific guidances** for generic drugs, on the FDA website; **21 CFR Part 320** (bioavailability and bioequivalence requirements)

FDA Bioavailability Studies Submitted in NDAs or INDs: General Considerations (draft, 2022): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-studies-submitted-ndas-or-ind-s-general-considerations>

ICH M12 Drug Interaction Studies (2024; published by EMA and FDA): <https://www.ema.europa.eu/en/ich-m12-drug-interaction-studies-scientific-guideline>

EMA Guideline on the pharmacokinetic and clinical evaluation of modified release dosage forms (EMA/CHMP/EWP/280/96 Rev1, 2014)

ICH E6(R3) Good Clinical Practice (2025), and the **EMA Guideline on computerised systems and electronic data in clinical trials** (EMA/INS/GCP/112288/2023)

CDISC SDTM Controlled Terminology (release 2026-03-27, source of the parameter codes): <https://evs.nci.nih.gov/ftp1/CDISC/SDTM/>

Appendices

Appendix A: Output Parameters

Every column the NCA can produce, with its label in the app, its definition and its unit for a dose in mg, time in h and concentrations in ng/mL. The unit follows the units you select. "With λ_z " means the parameter needs an accepted terminal fit; "single dose" means it is left empty at steady state. Partial AUC columns (Partial AUC t1–t2, Cmax t1–t2, Tmax t1–t2) have the unit of AUC, concentration and time. CL/F, Vz/F and the bioequivalence comparison of AUC to infinity use the observed AUC0– ∞ .

Code	Label in the app	Definition	Unit (example)	Reported for
b0	Y-Intercept (ln scale)	Intercept of the terminal regression on the ln scale	(none)	All, with λ_z
C _{MAX}	Peak Concentration (C _{max})	Highest observed concentration	ng/mL	All
T _{MAX}	Time of Peak (T _{max})	Time of C _{max}	h	All
T _{LAG}	Lag Time	Last sampling time before the first measured concentration	h	Extravascular
CL _{ST}	Last Measurable Concentration	Last measurable concentration (C _{last})	ng/mL	All
CL _{STP}	Predicted Last Concentration	C _{last} predicted by the terminal regression	ng/mL	All, with λ_z
T _{LST}	Time of Last Measurable Conc	Time of C _{last} (t)	h	All
LAM _{ZHL}	Half-Life	Terminal half-life, $\ln 2/\lambda_z$	h	All, with λ_z
LAM _Z	Elimination Rate Constant	Terminal elimination rate constant λ_z	1/h	All, with λ_z
LAM _{ZLL}	Lambda_z Lower Time	First time point of the terminal regression	h	All, with λ_z
LAM _{ZUL}	Lambda_z Upper Time	Last time point of the terminal regression	h	All, with λ_z
LAM _{ZNPT}	Points Used for Half-Life	Number of points in the terminal regression	(none)	All, with λ_z
CORRXY	Correlation (r)	Correlation coefficient of the terminal regression	(none)	All, with λ_z

R2	R-squared	R ² of the terminal regression	(none)	All, with λ_z
R2ADJ	Adjusted R-squared	Adjusted R ² of the terminal regression; the minimum applies to it	(none)	All, with λ_z
AUCLST	AUC to Last Point	AUC from 0 to Tlast (AUC0–t)	ng/mL·h	All
AUCALL	AUC All (incl. trailing zero)	AUC from 0 to the last sample, including a trailing zero	ng/mL·h	All
AUCIFO	AUC to Infinity (observed)	AUC0–t + Clast/ λ_z	ng/mL·h	Single dose, with λ_z
AUCIFP	AUC to Infinity (predicted)	AUC0–t + Clast,pred/ λ_z	ng/mL·h	Single dose, with λ_z
AUCPEO	AUC % Extrapolated (observed)	Share of AUC0– ∞ (observed) that is extrapolated, %	(none)	Single dose, with λ_z
AUCPEP	AUC % Extrapolated (predicted)	Share of AUC0– ∞ (predicted) that is extrapolated, %	(none)	Single dose, with λ_z
AUMCLST	AUMC to Last Point	Area under the first-moment curve ($C \times t$) from 0 to Tlast	ng/mL·h ²	All
AUMCIFO	AUMC to Infinity (observed)	AUMC extrapolated to infinity with Clast	ng/mL·h ²	Single dose, with λ_z
AUMCIFP	AUMC to Infinity (predicted)	AUMC extrapolated to infinity with Clast,pred	ng/mL·h ²	Single dose, with λ_z
AUMCPEO	AUMC % Extrapolated (obs)	Share of AUMC0– ∞ (observed) that is extrapolated, %	(none)	Single dose, with λ_z
AUMCPEP	AUMC % Extrapolated (pred)	Share of AUMC0– ∞ (predicted) that is extrapolated, %	(none)	Single dose, with λ_z
VZFO	Apparent Volume (Vz/F)	Dose/($\lambda_z \times \text{AUC0–}\infty$); at steady state from AUC _T	L	Extravascular, with λ_z
VZFP	Apparent Volume (pred)	As Vz/F, with the predicted AUC0– ∞	L	Extravascular single dose
CLFO	Apparent Clearance (CL/F)	Dose/AUC0– ∞ ; at steady state Dose/AUC _T	L/h	Extravascular

CLFP	Apparent Clearance (pred)	As CL/F, with the predicted $AUC_{0-\infty}$	L/h	Extravascular single dose
MRTEVLST	Mean Residence Time (to last)	$AUMC_{0-t}/AUC_{0-t}$, including absorption time	h	Extravascular single dose
MRTEVIFO	Mean Residence Time (to inf, obs)	$AUMC_{0-\infty}/AUC_{0-\infty}$ (observed), including absorption time	h	Extravascular single dose
MRTEVIFP	Mean Residence Time (to inf, pred)	As above, with the predicted values	h	Extravascular single dose
CMAX_DN	Dose-Normalised Cmax	Cmax divided by the dose of that profile	ng/mL per mg	With dose normalisation
AUCLST_DN	Dose-Normalised AUC Last	AUC_{0-t} divided by the dose	ng/mL·h per mg	With dose normalisation
AUCIFO_DN	Dose-Normalised AUC Inf	$AUC_{0-\infty}$ (observed) divided by the dose	ng/mL·h per mg	With dose normalisation
AUCIFP_DN	Dose-Normalised AUC Inf (pred)	$AUC_{0-\infty}$ (predicted) divided by the dose	ng/mL·h per mg	With dose normalisation
AUMCLST_DN	Dose-Normalised AUMC Last	$AUMC_{0-t}$ divided by the dose	ng/mL·h ² per mg	With dose normalisation
AUMCIFO_DN	Dose-Normalised AUMC Inf	$AUMC_{0-\infty}$ (observed) divided by the dose	ng/mL·h ² per mg	With dose normalisation
AUMCIFP_DN	Dose-Normalised AUMC Inf (pred)	$AUMC_{0-\infty}$ (predicted) divided by the dose	ng/mL·h ² per mg	With dose normalisation
AUCTAU	AUC Within Dosing Interval	AUC from 0 to τ ; extrapolated with λ_z when the last sample is before τ	ng/mL·h	Steady state
TAU	Dosing Interval (tau)	The dosing interval you entered	h	Steady state
CAVG	Average Concentration (Cavg)	AUC/τ	ng/mL	Steady state
CMIN_SS	Minimum Concentration (Cmin)	Lowest observed concentration from 0 to τ , pre-dose included	ng/mL	Steady state

CTAU_SS	Concentration at Tau (Ctau)	Observed concentration at τ ; empty without a sample at τ	ng/mL	Steady state
FLUCTP	Peak-Trough Fluctuation (%)	$(C_{\max} - C_{\min})/C_{\text{avg}} \times 100$	(none)	Steady state
SWING	Swing $((C_{\max} - C_{\min})/C_{\min})$	$(C_{\max} - C_{\min})/C_{\min}$	(none)	Steady state
C0	Initial Concentration (C0)	Concentration at time 0, back-extrapolated from the first two samples	ng/mL	IV bolus
AUCPBEO	AUC % Back-Extrapolated (observed)	Share of $AUC_{0-\infty}$ (observed) between 0 and the first sample, %	(none)	IV bolus single dose
AUCPBEP	AUC % Back-Extrapolated (predicted)	As above, with the predicted $AUC_{0-\infty}$	(none)	IV bolus single dose
VZO	Volume of Distribution (Vz)	Dose/ $(\lambda_z \times AUC_{0-\infty})$; at steady state from AUC_{τ}	L	IV, with λ_z
VZP	Volume of Distribution (pred)	As Vz, with the predicted $AUC_{0-\infty}$	L	IV single dose
CLO	Clearance (CL)	Dose/ $AUC_{0-\infty}$; at steady state Dose/ AUC_{τ}	L/h	IV
CLP	Clearance (pred)	As CL, with the predicted $AUC_{0-\infty}$	L/h	IV single dose
MRTIVLST	Mean Residence Time IV (to last)	$AUMC_{0-t}/AUC_{0-t}$ (minus half the infusion time)	h	IV single dose
MRTIVIFO	Mean Residence Time IV (to inf, obs)	$AUMC_{0-\infty}/AUC_{0-\infty}$ (observed; minus half the infusion time)	h	IV single dose
MRTIVIFP	Mean Residence Time IV (to inf, pred)	As above, with the predicted values	h	IV single dose
VSSO	Volume at Steady State (Vss)	Volume at steady state, $MRT \times CL$	L	IV single dose
VSSP	Volume at Steady State (pred)	As Vss, with the predicted values	L	IV single dose

Appendix B: Glossary

Term	Definition
ABE	Average bioequivalence (80–125% limits by default)
ABEL	Average bioequivalence with expanding limits (EMA, for highly variable drugs)
Adjusted R ²	Goodness of fit of the terminal regression, corrected for the number of points; the minimum in the app applies to it
ADNCA	CDISC ADaM dataset for non-compartmental analysis
ANOVA	Analysis of variance
AUC	Area under the concentration–time curve
AUC % extrapolated	The share of AUC to infinity that lies beyond the last measurable concentration
AUC to infinity (AUC _{0–∞})	AUC to last point plus C _{last} /λ _z
AUC to last point (AUC _{0–t})	AUC from 0 to the last measurable concentration
AUC _τ	AUC within the dosing interval (steady state)
BLQ	Below the limit of quantification
CL/F	Apparent clearance: clearance divided by the bioavailability F
C _{max}	Maximum observed concentration
Crossover	A design in which each subject receives the treatments in turn, in periods separated by a washout
CV%	Coefficient of variation
CV _{wR} / CV _{wT}	Within-subject coefficient of variation of the Reference / Test
DDI	Drug–drug interaction
EMA Method A / B	The fixed-effects (A) and mixed-effects (B) ANOVA of the EMA questions and answers on bioequivalence
Extravascular	Given other than into a vein (oral, intramuscular, subcutaneous), so the drug is absorbed first
Geometric mean	exp(mean of the ln values); the usual summary for AUC and C _{max}
Geometric CV	$100 \cdot \sqrt{(\exp(\text{SD}^2 \text{ of the ln values}) - 1)}$
GMR	Geometric mean ratio (Test/Reference)
HVD	Highly variable drug (within-subject CV of the Reference above 30%)
ICH M13A	ICH guideline on bioequivalence for immediate-release solid oral dosage forms (2024)
Intersection-union test	A test that passes only if each of several tests passes; no multiplicity correction is needed

λ_z	Terminal elimination rate constant
LLOQ	Lower limit of quantification
MRT	Mean residence time
MSE	Mean squared error: the residual variance of the ANOVA model
NCA	Non-compartmental analysis
NTID	Narrow therapeutic index drug
Owen's Q	The function PowerTOST uses for exact power of the two one-sided tests
Partial AUC	AUC over part of the profile, for example AUC 0–30 min; the interval comes from the protocol or a product-specific guidance
PE	Point estimate (geometric mean ratio)
Period	One administration in a crossover, with its own profile
Pivotal metric	A bioequivalence metric that has to meet the acceptance limits; it receives a verdict
Power	The probability that a study shows bioequivalence when the products are bioequivalent as assumed
Precipitant (perpetrator)	The drug that causes a drug interaction
Profile	The concentrations of one subject under one treatment in one period
Reference	The treatment in the denominator of the ratio, chosen by the user
RSABE	Reference-scaled average bioequivalence (FDA, for highly variable drugs)
Sequence	The order in which a subject receives the treatments, for example TR or RT
Supportive metric	A bioequivalence metric reported with its ratio and confidence interval, without a verdict
t (last measurable)	The time of the last measurable concentration of a profile; an interval written as 4 h–t ends there
$t_{1/2}$	Terminal elimination half-life
T _{max}	Time of the maximum observed concentration
TOST	Two one-sided tests
Victim	The drug whose pharmacokinetics a drug interaction changes
V _z /F	Apparent terminal volume of distribution: the volume divided by F
Washout	The time between periods of a crossover, long enough for the drug to disappear
Welch interval	A confidence interval for a parallel design that does not assume equal variances in the two groups

In this manual, "missing" and "excluded" for a BLQ value mean the same: the value is left out of the calculation.

Appendix C: Recognised Column Names

Meaning	Recognised names (case-insensitive)
Subject	Subject, SubjectID, USUBJID, ID, Pat, PatID, Proband, Teilnehmer
Time	TAD (preferred over TIME), Time, Hours, Hour, Tpt, NTIM, APTS, Zeit, Tid
Concentration	Conc, Concentration, DV, Cp, Result, Konz, Plasma, or a name containing ng/mL or ug/L
Treatment	Treatment, Trt, Formulation, Drug, Arm, Behandlung
Period	Period, Per, Prd, Phase, APERIOD, Occasion, OCC (not Visit)
Sequence	Sequence, Seq, Grp
Dose	Dose, Amt, Amount, Dosis

When no name matches, the app falls back to a column by position, so always check the suggested mapping. When the file has another column that could be the time or the concentration, a note says which one was chosen and names the other.

Appendix D: Troubleshooting

Opening and Installing

“Disconnected from the server”: the public app was idle too long, the analysis exceeded memory or time limits, or the app met an unexpected error. Reload the page, upload the file again and click Process Data; results that were not downloaded are lost. If it happens again, run the app locally and check the console.

The installation stops with an error: see Installing the App on Your Own Computer.

Uploading and Processing

Comma decimals not recognised: set Decimal point to comma (and the Delimiter, often semicolon) in the upload settings.

A decimal point in a decimal-comma file: the file mixes decimal marks. Correct the values, or set the decimal mark to period if the file uses points throughout.

“N BLQ entries detected but LLOQ is not set”: set the LLOQ and click Process Data again; for “<X” entries, the button Set LLOQ to X fills in the value (Tutorial 2).

“Profile(s) do not start near time zero”: time is probably measured from the first dose of the study. Use the time since the dose of each period (in CDISC terms ARRLT or NRRLT, not AFRLT).

Subject IDs that restart in each sequence: give every subject an ID that is unique in the study, for example TR-01 and RT-01.

“The data give concentrations in ...”: the unit column of the file says one unit and the analysis settings another. Select the unit the file states, or correct the file.

Large sampling gap warning: for studies longer than 48 h the threshold is 48 h, so daily trough samples are not flagged; for shorter studies it is 24 h. A gap close to the threshold is usually normal sampling.

“Refused: ...” in the CDISC ADNCA upload: the message says why (for example derived records, several units, AFRLT as time, or AVAL missing without a BLQ result). Fix the dataset upstream or make the choice the message asks for.

NCA

Unit error before the NCA runs: the combination of dose, time and concentration units is not supported, or molar units are combined with a mass dose without a molecular weight. Select supported units and enter the molecular weight.

“NCA failed”: the message gives the reason, usually no profile with at least 2 positive concentrations, or a mapping problem. Check the mapping and the quality report.

Half-life is empty: the automatic fit is below the minimum adjusted R^2 , or fewer than 3 usable points remain. Review the profile in Half-Life Review, or lower the minimum if your analysis plan allows it.

Half-life differs from other NCA software: check the minimum R^2 , the point selection in Half-Life Review, the BLQ rule and the trapezoidal method; programs differ in how they choose the terminal-phase points.

My partial AUC is empty: the interval reaches past the last measurable concentration of that profile, or starts after it. The note above the results names the profile and its last measurable time. Partial AUCs are not extrapolated.

Bioequivalence

“Choose the Reference treatment”: the run is blocked until you choose the Reference treatment in the Bioequivalence settings. The app pre-selects it only for unambiguous names (R, Ref, Reference, Comparator, Innovator, Originator, RLD).

No verdict: see When There Is No Verdict, in Path 6. The Bioequivalent? column gives the reason.

No verdict for my partial AUC: the interval is marked supportive, or the metric has a value of zero in at least one profile. The table gives the reason and the number of zero values per treatment.

“Treatment column must have exactly 2 levels”: see Studies with Three or More Treatments.

Figures

Download Figure is grey: it becomes active once the first plot has appeared; wait for it.

Time points are missing from my summary plot: the summary plot averages per time value. With actual sampling times (0.95 h and 1.05 h for a nominal 1 h), use a copy of the file with nominal times for the figure; keep the actual times for the NCA.

“N observations with concentration ≤ 0 excluded”: zeros, for example BLQ values set to 0 by Rules 1, 2 or 5, cannot be log-transformed, so they are left out of the geometric mean at that time point. They stay in your data.

Too many colours to distinguish: try the Colorblind-safe (Okabe-Ito) palette. With more than eight groups, switch to Grayscale and annotate the exported figure in a graphics program.

Records and Exports

The reproduction check says DIFFERENT: a file in the record was changed after it was made, a parameter is present on one side only, or the results differ. `reproduction_check.txt` gives the reason.

The record holds a CSV instead of my Excel file: the uploaded file was no longer available; see Complete Analysis Record.

Appendix E: References

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Appendix F: Document Control

Document ID NCA-UM-001. The PDF of this manual is part of each release of the app; its SHA-256 is listed in validation/release_manifest.csv, so an installation can be checked to hold the manual of its release.

Version	Date	Author	Changes
1.0	March 2026	Rob ter Heine	Initial release
1.1	April 2026	Rob ter Heine	Added theoretical background (Part V), expanded statistical methods (Part VI), DDI study analysis, decimal separator guidance
1.2	April 2026	Rob ter Heine	Quick Start data preparation callout, grid plot fix documentation, single-subject Analysis Record, DDI interpretation thresholds, regulatory DDI guideline links, troubleshooting updates
1.3	April 2026	Rob ter Heine	New chapter 3 (Visualize Data); chapters 3–5 renumbered to 4–6; hub layout updated to 3+3 grid; attribution updated
1.4	April 2026	Rob ter Heine	Factual corrections (Rule 6 label, AUC extrapolation flag); new feature documentation (steady-state in BE, balance check, design summary, Apply LLOQ button, excluded profiles note); beginner clarity additions (LLOQ/BLQ definitions, extravascular explained, R ² failure consequences, SHA-256 explained, half-life review how-to, AUC % Extrapolated meaning)

1.5	September 2026	Rob ter Heine	Updated for NCA Assistant v1.4.0: CDISC ADNCA upload and parameter codes, safety checks on upload, decimal comma and semicolon files, doses per period, replicate designs and the within-subject variability table, choosing the Reference treatment, the minimum R^2 rule, planning with Test and Reference CV, the reproduction check in the Analysis Record. Exercise 1 answers corrected; guideline references updated.
1.6	September 2026	Rob ter Heine	Updated for NCA Assistant v1.5.0: partial AUCs in Paths 4, 5 and 6, with intervals ending at a time or at the last measurable concentration, Cmax and Tmax within an interval, pivotal and supportive metrics in bioequivalence, and interval shading in Visualize Data. New Exercise 4, new sections in Parts V and VI, and new glossary, troubleshooting and reference entries.
1.7	September 2026	Rob ter Heine	Updated for NCA Assistant v1.6.0, after a review of the whole app. New: intended use and data notice; ICH M13A (Cmax and AUC to last point compared by default, BLQ values kept out of the half-life, checks listed with the results, no verdict where the guideline excludes one); widened limits for Cmax only; Rules 4 and 6 changed; IV bolus samples at time 0 set aside; thousands separators in decimal-comma files; unique subject IDs; units taken from the data; settings kept and results cleared; file integrity in the reproduction check; the Welch interval; excluding subjects. Exercise answers checked against v1.6.0.
1.8	25 September 2026	Rob ter Heine	For NCA Assistant v1.7.0, after a review by a tutor, a clinical pharmacologist and a regulatory compliance officer. New: Getting Started with local installation, seven tutorials with an answer key, study types (IV bolus and infusion, single and

			multiple ascending dose, food effect, metabolites, bioavailability, three or more treatments), Reading Your Results, a chapter on regulated use, derived-parameter formulas, an appendix of output parameters, and troubleshooting by stage. Corrected throughout for the app changes of v1.7.0.
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